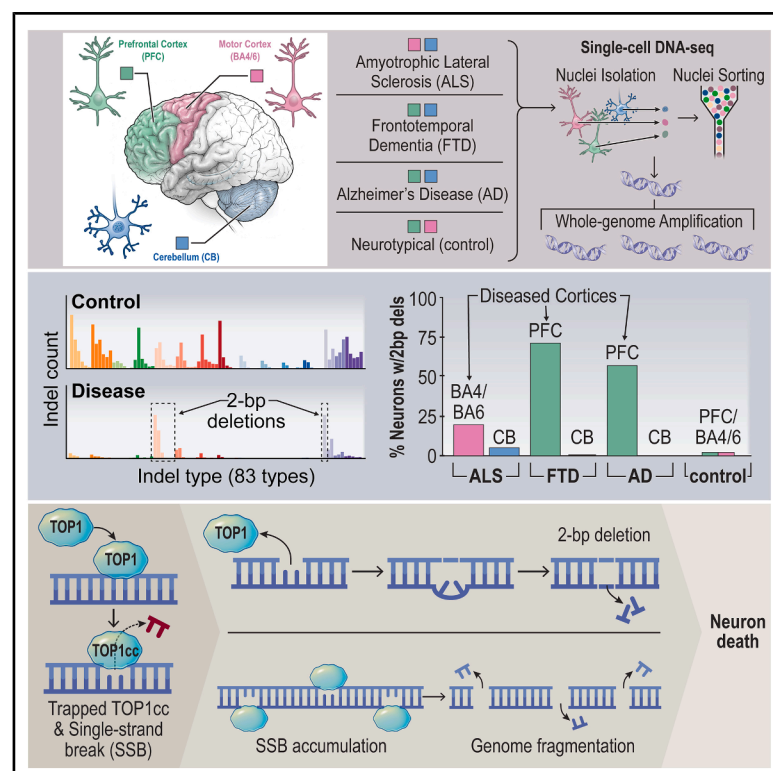


Recurrent patterns of TOP1-mediated neuronal genomic damage shared by major neurodegenerative disorders

Graphical abstract



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In brief

Single-cell whole-genome sequencing reveals a shared pattern of excessive somatic mutations across *C9ORF72* ALS, *C9ORF72* FTD, and Alzheimer's disease, dominated by 2-bp deletions. These mutations arise from aberrant topoisomerase 1 (TOP1)-mediated mutagenesis linked to oxidative DNA damage, identifying a unifying mechanism of neuronal genomic instability in neurodegeneration.

Highlights

- Neurons from ALS, FTD, and Alzheimer's disease show increased sSNVs and sIndels
- A shared mutational signature of 2-bp deletions reflects TOP1-mediated mutagenesis
- TOP1 cleavage complexes and single-strand DNA breaks are elevated in diseased neurons
- TOP1-mediated genomic damage is absent in cerebellar granule cells

Article

Recurrent patterns of TOP1-mediated neuronal genomic damage shared by major neurodegenerative disorders

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SUMMARY

Amyotrophic lateral sclerosis (ALS), frontotemporal dementia (FTD), and Alzheimer's disease (AD) represent two major categories of neurodegenerative disorders—TAR DNA-binding protein 43 (TDP-43) and tau proteinopathies—for which the mechanisms driving neuronal death remain unclear. Single-cell whole-genome sequencing of 469 neurons from *C9ORF72* ALS, *C9ORF72* FTD, AD, and control brains revealed increased somatic single-nucleotide variants (sSNVs) and insertions/deletions (sIndels) in all three diseases. Mutational signature analysis identified a disease-associated sSNV signature consistent with oxidative damage and an sIndel process affecting 22% of ALS, 76% of FTD, and 61% of AD neurons—but only 2% of control neurons—resembling signature ID4, previously linked to topoisomerase 1 (TOP1)-mediated mutagenesis. Rapid approach to DNA adduct recovery (RADAR) assays confirmed increased TOP1-DNA covalent complexes, and duplex sequencing confirmed the increased sIndels and identified single-strand events as likely precursor lesions. TOP1-associated sIndel mutagenesis and genome instability thus represent a mechanism shared by both TDP-43 and tau neurodegeneration.

INTRODUCTION

Amyotrophic lateral sclerosis (ALS), frontotemporal dementia (FTD), and Alzheimer's disease (AD)^{1,2} are common neurodegenerative diseases with diverse pathologies and genetic underpinnings. ALS and FTD are closely related, with clinical, pathological, and genetic overlap, marked by the depletion of nuclear TAR DNA-binding protein 43 (TDP-43) and the accumulation of cytoplasmic TDP-43 inclusions in neurons. Mutations in many

genes, such as *C9ORF72*, *TARDBP* (TDP-43), *FUS*, *TBK1*, *VCP*, and *SQSTM1*, contribute to both diseases,³ with *C9ORF72* repeat expansion being the most common genetic cause of both familial and sporadic forms of ALS and FTD.^{4,5} By contrast, AD pathology is characterized by the accumulation of amyloid- β protein and phosphorylated tau and prominent activation of microglia and shows distinct underlying genetic risks.⁶

Genomic instability, which leads to the accumulation of DNA damage and mutations, has been implicated in all of these

conditions, but its exact role remains elusive. In ALS, FTD, and AD, increased DNA damage due to high levels of oxidative stress, R-loops, and DNA strand breaks has been reported.^{7–16} Several ALS/FTD-related genes are involved in DNA damage responses, such as *SOD1*, *FUS*, *TDP-43*, *C9ORF72*, *NEK1*, *SQSTM1*, *SETX*, and *VCP*.^{9,16–24} *C9ORF72* repeat expansions have recently been shown to induce chromosomal fragility and DNA damage.²⁵ Neurons are particularly vulnerable to DNA damage due to their high transcriptional activity, large energy demand, and associated oxidative stress. Recent studies have shown that somatic single-nucleotide variants (sSNVs) accumulate linearly with age in neurons, with further increases observed in neurodegenerative conditions, including AD, Cockayne syndrome, and Xeroderma pigmentosum.^{13,14,26}

In this study, we compared somatic mutations in neurons from major neurodegenerative diseases associated with TDP-43 proteinopathy (ALS and FTD) and tau proteinopathy (AD) to compare potential mechanisms of mutagenesis between these two major categories of neurodegenerative diseases. We profiled sSNVs and somatic insertions and deletions (sIndels) in neurons with and without depletion of nuclear TDP-43 from the motor or premotor cortex of six *C9ORF72* ALS and the prefrontal cortex of six *C9ORF72* FTD brains using single-cell whole-genome sequencing (scWGS). We then compared the results to those from AD neurons,¹⁴ in which sIndels were not analyzed and neurons from neurotypical controls. Our analysis revealed significantly elevated levels of sSNVs and sIndels in neurons from *C9ORF72* ALS, *C9ORF72* FTD, and AD brains, but in surprisingly similar patterns despite their diverse genetic basis and pathologies. Importantly, our findings indicate that neurons in all three disease conditions often exhibit an excessive number of sIndels—sometimes over 1,000, equivalent to hundreds of years of age-related sIndel accumulation—and share a mutational pattern characterized by two-base-pair (2-bp) deletions. Computational and experimental analyses of these sIndels associate the mutagenic process with topoisomerase 1 (TOP1)-mediated mutagenesis and show that this mutagenic process is absent in cerebellar neurons from the same diseased brains. Our results suggest that increased DNA damage and accelerated somatic mutation contribute to the pathogenesis of degenerative disorders with diverse pathological features.

RESULTS

Isolation of neuronal nuclei with TDP-43 pathology from *C9ORF72* ALS and FTD brains achieves high purity

To isolate neuronal nuclei with either normal or depleted nuclear TDP-43 (TDP-43+ and TDP-43–) from postmortem *C9ORF72* ALS and FTD brains, we performed fluorescence-activated nuclear sorting (FANS) with co-staining of NeuN and TDP-43 antibodies²⁷ (Figure 1A). TDP-43– neurons were only found in *C9ORF72* ALS and FTD brains and were absent in age-matched controls (Figure 1B). Single-nucleus RNA sequencing (snRNA-seq) data from isolated nuclei revealed that 96.4% of TDP-43+/NeuN+ and 95.4% of TDP-43–/NeuN+ nuclei were indeed neuronal (Figures 1C and S1A). Furthermore, we examined cryptic exons caused by loss of TDP-43 function during RNA splicing in the snRNA-seq data.^{28–30} Transcripts containing

known cryptic exons in the *RAP1GAP*, *STMN2*, *ATP8A2*, and *KALRN* genes were detected in TDP-43– neurons but were absent or very rare in TDP-43+ neurons (Figures 1D, 1E, S1B, and S1C), confirming the presence of TDP-43 disease pathology in isolated TDP-43– neurons.

Both TDP-43+ and TDP-43– neurons encompassed all major clusters of excitatory and inhibitory neurons (Figure 1C), though inhibitory neurons were notably depleted in the TDP-43– population (Figures 1C and S1D), confirmed by the reduced expression of inhibitory neuron markers in TDP-43– neurons compared with TDP-43+ neurons in previously generated bulk RNA-seq data from *C9ORF72* FTD brains (Figure S1E).²⁷ These results are consistent with recent studies demonstrating that inhibitory neurons rarely exhibit cytoplasmic TDP-43 inclusions.³¹ However, it remains unclear whether the reduced TDP-43 pathology in inhibitory neurons reflects resistance or susceptibility to TDP-43 pathology, given the observed loss of inhibitory neurons in ALS.^{32–34}

Somatic mutation burdens are increased in diseased neurons

We performed primary template-directed amplification (PTA)-based single-cell whole-genome amplification (scWGA) and scWGS on neurons isolated from the motor cortex or premotor cortex of six *C9ORF72* ALS brains and the prefrontal cortex of six *C9ORF72* FTD brains (three TDP-43+ and three TDP-43– neurons per brain; Tables S1 and S2) and identified sSNVs and sIndels using single-cell analysis 2 (SCAN2),²⁶ our recently developed single-cell genotyper (Figure 1A). Somatic mutations in these diseased neurons were compared with those in 24 newly sequenced premotor cortex neurons and 56 previously sequenced prefrontal cortex neurons from neurotypical controls across various ages,^{26,35} as well as 29 previously sequenced prefrontal cortex neurons from AD brains (Figure 1A; Tables S1–S3).¹⁴ Cells that did not pass quality control (QC) criteria, such as amplification uniformity,³⁶ were excluded from downstream analyses (Figures S2 and S3; Table S2; see STAR Methods). SCAN2 recovered approximately 35% of sSNVs and 24% of sIndels, producing a somatic mutation catalog with >67,000 sSNVs and >12,000 sIndels with an aggregate estimated error rate of <10% (Figure S2; Table S4).

Many neurons from *C9ORF72* ALS, *C9ORF72* FTD, and AD brains exhibited elevated burdens of both sSNVs and sIndels, with sIndels showing greater increases (Figure 2A; Table S5). Substantial variability in sSNV and sIndel burdens was observed between neurons from brains with the same disease and, in some cases, even within the same brain (Data S1). Since age-matched neurons from the prefrontal and premotor cortex of neurotypical controls aged >50 years showed similar burdens of sSNVs and sIndels (Figure 2B), neurons from these two sites were combined in all subsequent analyses. After adjusting for age, burdens of both sSNVs and sIndels in neurons from FTD and AD were significantly increased compared with those from neurotypical controls (Figure 2C). In ALS neurons, sIndel burdens were significantly increased but not sSNV burdens (Figure 2C). The relatively modest increase in somatic mutations in *C9ORF72* ALS neurons compared with *C9ORF72* FTD and AD neurons may reflect the selective involvement of motor neurons

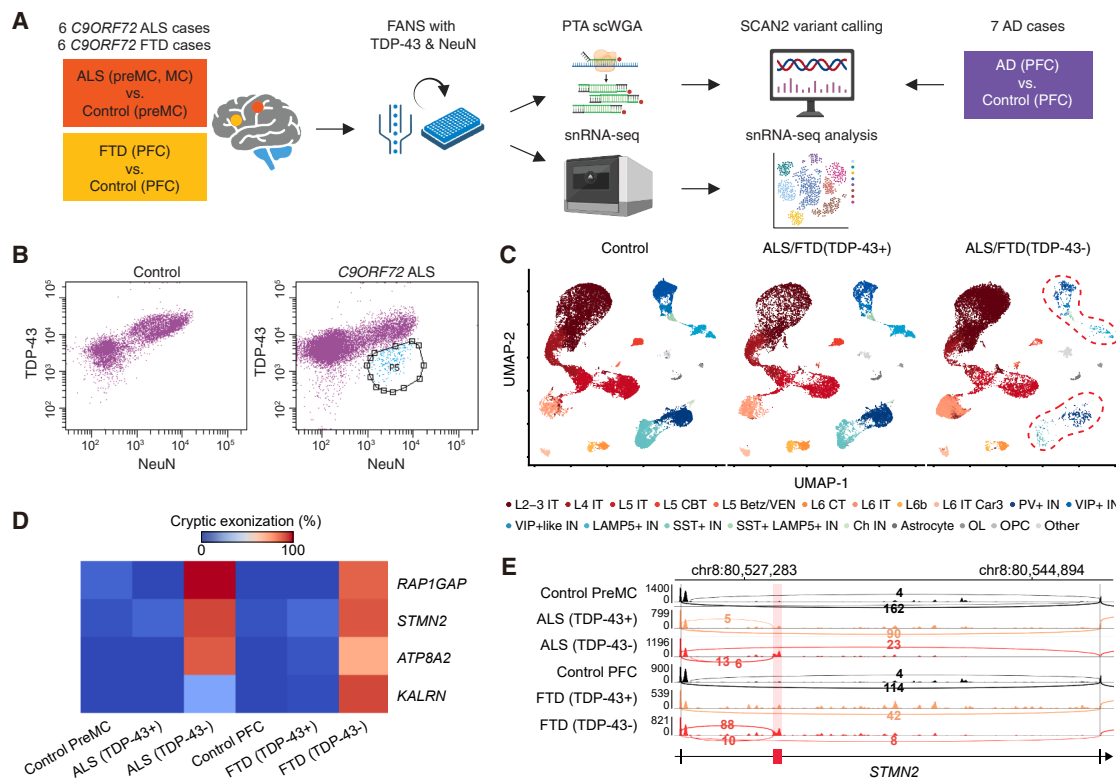


Figure 1. Experimental strategy and characterization of isolated neuronal nuclei from *C9ORF72* ALS and FTD brains

(A) Overview of the experimental design. Neuronal nuclei with and without nuclear TDP-43 depletion were isolated from 6 *C9ORF72* ALS and 6 *C9ORF72* FTD brains, then subjected to genome sequencing and analysis. These were compared with neuronal nuclei from neurotypical brains, including newly generated data and previously published datasets. Previously generated scWGS data from AD brains were included as well.

(B) Representative FACS plots showing neuronal nuclei from a *C9ORF72* ALS brain and a neurotypical control brain co-stained with NeuN and TDP-43 antibodies. The P5 gate (right, circle) indicates neuronal nuclei with depletion of nuclear TDP-43.

(C) UMAP clustering of snRNA-seq data from neuronal nuclei, colored by major cell types. TDP-43[−] neuronal nuclei come exclusively from *C9ORF72* ALS and FTD brains, while TDP-43⁺ neuronal nuclei come from both diseased and neurotypical brains, and control indicates neurotypical brains. IT, intratelencephalic neurons; CBT, corticobulbar tract neurons; VEN, Von Economo neurons; CT, corticothalamic neurons; PV, parvalbumin; IN, inhibitory neurons; Ch IN, cholinergic inhibitory neurons; OL, oligodendrocytes; OPC, oligodendrocyte precursor cells. Red dashed circles emphasize reduced inhibitory neurons under the TDP-43[−] condition.

(D) Heatmap of cryptic exonization showing the proportion of transcripts with cryptic exons in neurons across different conditions.

(E) IGV genome browser tracks of *STMN2* showing snRNA-seq read coverage and splice junctions. TDP-43[−] neurons (red) contain a marked increase in cryptic exon inclusion (highlighted region), which is absent or low in TDP-43⁺ (orange) and control (black) neurons. Numbers next to splice junctions are read counts supporting the junction. PFC, prefrontal cortex; preMC, premotor cortex.

See also Figure S1.

in ALS and their rapid and almost complete disappearance by the time of death, whereas a broader range of neuronal types are involved in FTD and AD.³⁷ Together, our results show that accumulation of sSNVs and sIndels is accelerated in *C9ORF72* ALS, *C9ORF72* FTD, and AD neurons. Unexpectedly, TDP-43[−] neurons from *C9ORF72* ALS and FTD brains did not exhibit higher burdens of sSNVs and sIndels compared with TDP-43⁺ neurons (Figure 2D), despite TDP-43's involvement in DNA damage repair of double-strand breaks.⁹ Merging neurons from *C9ORF72* ALS and FTD cases to increase statistical power suggested a marginal increase in sIndels in TDP-43⁺ relative to TDP-43[−] neurons (Figure 2E), but this effect was small and may reflect differences in neuronal survival or sampling. Nonetheless, our data suggest that *C9ORF72* repeat expansion and depletion of nuclear TDP-43 are not synergistic in causing DNA damage.

A recurrent pattern of 2-bp deletions across diseased neurons

Among several patterns of mutations observed in our mutation catalog, we uncovered a disease-specific signature characterized by 2-bp deletions that was greatly elevated in neurons from all three disease conditions compared with controls (Figures 3A and 3B; Table S5). To investigate potential mutagenic mechanisms, we performed *de novo* mutational signature extraction, a technique that can identify underlying processes based on the types and frequencies of mutations.³⁸ *De novo* mutational signature extraction was performed on neuronal mutations from the three neurodegenerative conditions, along with (1) 81 neurons from AD, as well as 190 neurons and 40 glial cells from neurotypical controls previously amplified using multiple displacement amplification (MDA), an earlier whole-genome

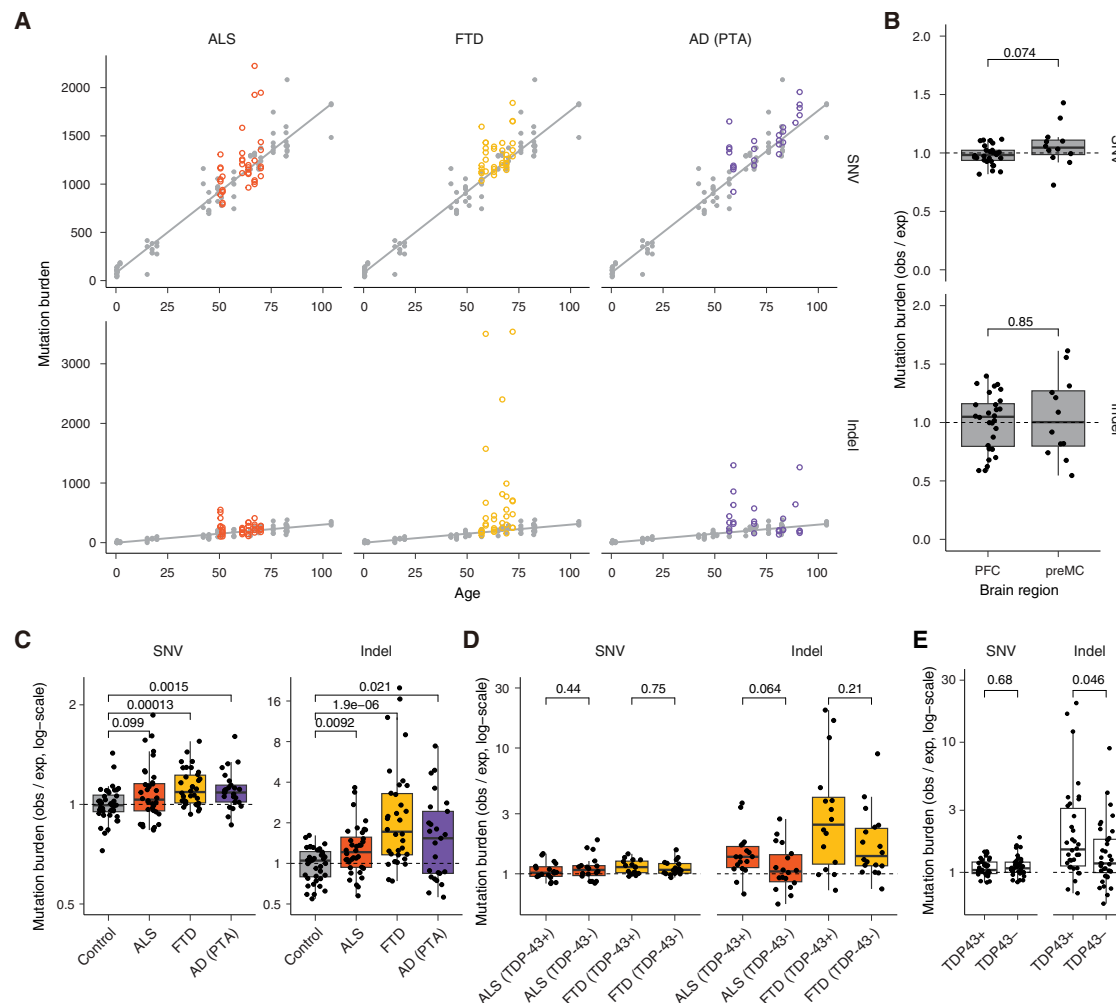


Figure 2. Increased somatic mutation burdens in neurons from C9ORF72 ALS, C9ORF72 FTD, and AD brains

(A) Extrapolated genome-wide sSNV and sIndel burdens for neurons from C9ORF72 ALS, C9ORF72 FTD, and AD brains compared with those for neurons from neurotypical brains as a function of age. Identical control neuron points and regression (see STAR Methods) are shown in gray in each panel for comparison.

(B) Burdens of sSNVs and sIndels in PFC (prefrontal cortex) and preMC (premotor cortex) neurons from neurotypical brains aged >50 years. Each point represents the burden of one neuron, compared against the expected burden for its age (see STAR Methods).

(C) Mutation burdens corrected for age (see STAR Methods). preMC and PFC neurons are combined to form the control set.

(D) TDP-43+ and TDP-43- neurons from C9ORF72 ALS and FTD brains show similar levels of sSNVs and sIndels.

(E) Neurons from ALS and FTD cases were combined into a single group of TDP-43+ and TDP-43- neurons.

All *p* values in this figure represent Wilcoxon rank sum tests.

See also Figures S2 and S3.

amplification method,³⁹ and (2) an additional 66 oligodendrocytes from neurotypical controls amplified using PTA.³⁵ This analysis resulted in six *de novo* SNV signatures (SBS-A to SBS-F, Figure S4A) of which only SBS-B showed an increase in PTA neurons from all three neurodegenerative conditions compared with control PTA neurons, although the increase in ALS neurons did not reach statistical significance (Figures S4B and S4C).

To explore the mutagenic processes associated with the *de novo* signatures, we compared them to existing mutational signatures in the Catalogue of Somatic Mutations in Cancer (COSMIC) database. SBS-B closely resembles the COSMIC

SBS30 signature (cosine similarity 0.90), which is associated with inactivating mutations in *NTHL1*, an enzyme involved in base excision repair of oxidative DNA damage.⁴⁰ The elevation of SBS-B sSNVs in neurons from neurodegenerative conditions may therefore indicate increased levels of oxidative DNA damage, consistent with previous studies revealing increased oxidative stress in these diseases.^{7,10,12} SBS-A and SBS-D resemble COSMIC SBS5 and SBS1, respectively, both of which are “clock-like” signatures. While SBS1 is thought to be cell-division dependent and thus limited in postmitotic neurons, SBS5 accumulates universally during aging. The mechanisms underlying SBS-C and SBS-E (which showed no increase in diseased

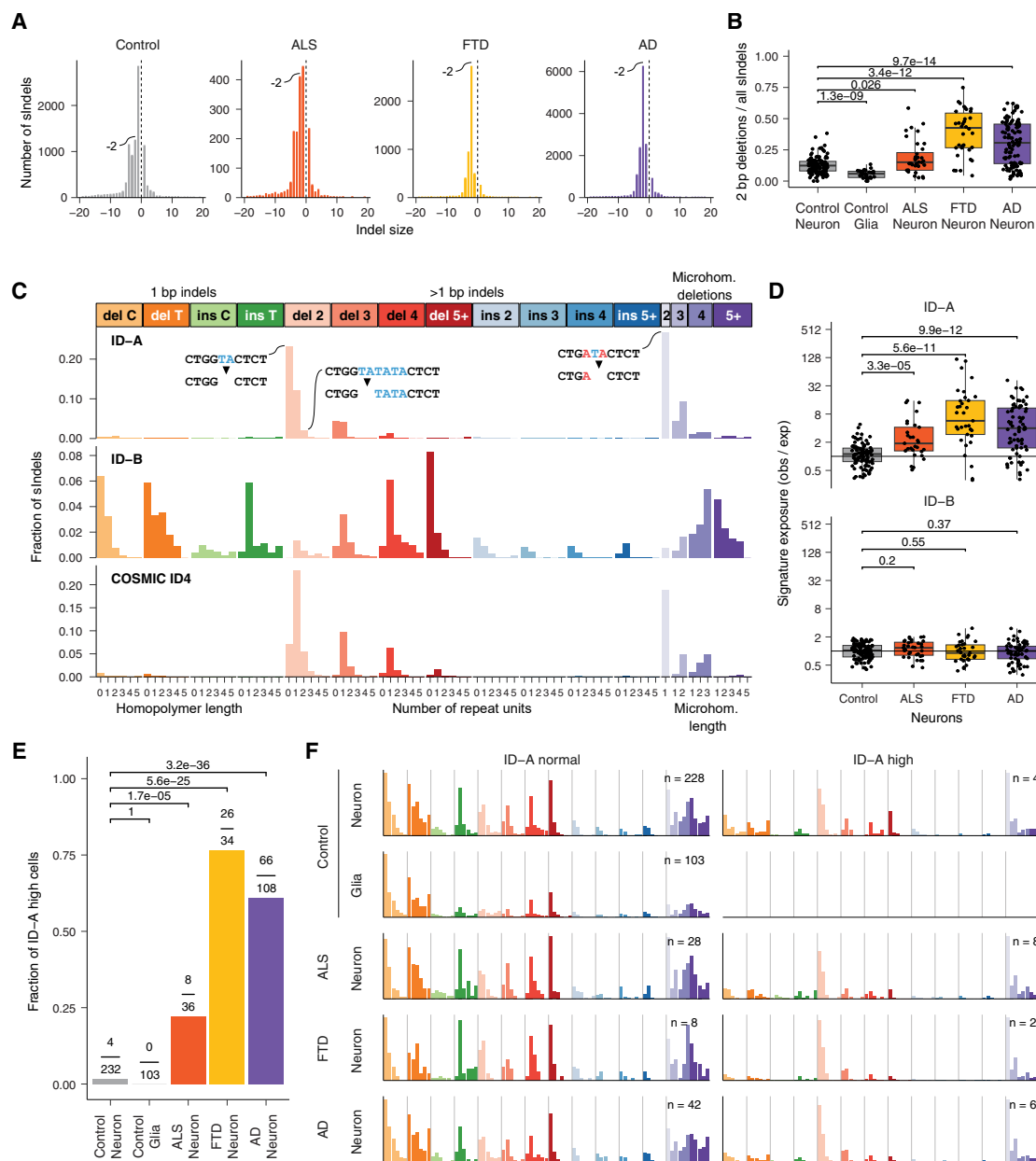


Figure 3. Neurons from *C9ORF72* ALS, *C9ORF72* FTD, and AD brains share an indel mutational signature

(A) Distribution of indel lengths across conditions. Diseased neurons have higher levels of 2-bp deletions compared with controls.

(B) Fraction of 2-bp deletions in neurons and glia across conditions. p values represent Wilcoxon rank-sum tests. Glia include pure oligodendrocytes and mixed glia.

(C) Two *de novo* indel signatures identified from joint analysis of diseased and neurotypical control neurons amplified by PTA, as well as previously published neurons and glia amplified using an older amplification technology (MDA). COSMIC ID4 is a signature previously identified in cancer and normal aging neurons. Three examples of deletions that would be classified into the corresponding bars are provided to illustrate the disease-associated 2-bp deletions in ID-A.

(D) Burdens of ID-A and ID-B indels. p values represent Wilcoxon rank-sum tests.

(E) Fraction of cells with high levels of ID-A indels in neurons and glia across conditions, determined by a classifier (see STAR Methods). p values represent Fisher tests of independence between disease status and ID-A-high classifications. Two neurons and three oligodendrocytes with 0 indels were excluded from classification.

(F) Indel spectra of neurons and glia stratified by ID-A-high and -normal classifications.

See also Figures S5 and S6

neurons) and SBS-F (which was increased in ALS and AD neurons) are unclear.

De novo mutational signature extraction of sIndels identified three distinct signatures, ID-A, ID-B, and ID-C (Figures 3C and S5A). ID-A burdens were significantly increased in PTA neurons from the three neurodegenerative conditions compared with control PTA neurons, while burdens of ID-B showed no difference across PTA neurons from all conditions (Figures 3D, S5B, and S5C). ID-C was elevated only in MDA-amplified AD neurons but not in PTA-amplified AD neurons ($p = 3.9 \times 10^{-4}$, Wilcoxon rank-sum test), suggesting that this signature may represent overfitting to MDA-specific artifacts (Figure S5C). ID-A is dominated by 2-bp deletions (Data S2) and most closely resembles the COSMIC signature ID4 (cosine similarity 0.80), which has recently been linked to deficiency in RNase H2, an enzyme responsible for removing ribonucleotides erroneously incorporated into the genome. In the absence of functional RNase H2, ribonucleotides can instead be removed through TOP1-mediated ribonucleotide excision repair (RER).⁴¹ Although ID4 is a feature of normal aging in neurotypical neurons—even more so than the canonical “clock-like” indel signatures ID5 and ID8 identified in cancers^{26,35,42}—the excessive ID-A burden observed in all three neurodegenerative conditions suggests a dramatic acceleration of this age-associated process. Indeed, virtually all of the excess sIndel burdens in diseased neurons can be attributed to ID-A.

We constructed a classifier (see STAR Methods) to determine which cells exhibited ID-A burdens in line with typical neuronal aging (ID-A normal) or an excessive ID-A burden (ID-A high). Upon stratifying our cells by ID-A status (Figure 3E), we found significantly higher fractions of ID-A-high neurons in every neurodegenerative condition (ALS: odds ratio [OR] = 15.1, $p = 1.68 \times 10^{-5}$; FTD: OR = 158.3, $p = 5.58 \times 10^{-25}$; AD: OR = 79.5, $p = 3.22 \times 10^{-36}$; all comparisons represent one-sided Fisher’s tests vs. control neurons) while only 4/232 control neurons and none of the 103 control glia were classified as ID-A high. Furthermore, 14 of 19 diseased brains contained at least one ID-A-high PTA-amplified neuron, and the mutational spectra of ID-A-high and -normal neurons were remarkably similar across conditions (Figure 3F), supporting the notion that ID-A pathology is likely a universal feature of these diverse neurodegenerative conditions rather than an isolated occurrence. Consistent with the overall burdens of sSNVs and sIndels, the levels of these *de novo* mutational signatures did not differ between TDP-43+ and TDP-43– neurons (Figure S6), further indicating that nuclear TDP-43 depletion does not enhance mutagenesis in individuals with *C9ORF72* expansions.

ID-A deletions exhibit features of TOP1-mediated mutagenesis

To further investigate the association between ID-A sIndels in diseased neurons and TOP1-mediated mutagenesis, we examined features of the 2-bp deletions. Mammalian TOP1 exhibits a preference for thymine bases and cleaves at the 3′ phosphodiester bond following the thymine nucleotide,⁴³ primarily causing deletions at a thymine-any nucleotide-thymine (TNT) motif, with the most common deletions being CT dinucleotides (Figure 4A).⁴¹ Consistent with this observation, the 2-bp dele-

tions in neurons from neurodegenerative conditions were enriched in CT deletions, and deletions with microhomology or at least 2 repeat units occurred most frequently within TNT motifs (Figures 4B and 4C).

Neurons from *C9ORF72* FTD and AD cases further showed increased levels of the TOP1 cleavage complex (TOP1cc), a protein complex formed through a covalent TOP1-DNA crosslink during TOP1-mediated RER (Figure 4A). We performed the rapid approach to DNA adduct recovery (RADAR) assay to measure TOP1cc levels directly in neuronal nuclei isolated from the prefrontal cortex of *C9ORF72* FTD, AD, and age-matched control brains.⁴⁴ TOP1cc signals were significantly increased in diseased neurons compared with controls (Figures 4D and 4E, see STAR Methods). In parallel, we analyzed neuronal nuclei that were positive for both TOP1cc and NeuN antibody staining from the prefrontal cortex of *C9ORF72* FTD, AD, and control brains using flow cytometry and found higher fractions of neurons with increased TOP1cc immunoreactivity in disease brains (2.8%–11.6%) compared with controls (0.2%–0.5%) (Figure S7A). These results provide direct evidence of increased formation of covalent TOP1 links to DNA in *C9ORF72* FTD and AD neurons.

TOP1-mediated ID4 indels can arise during both replication and transcription in mitotic cells⁴⁵ and are enriched at transcribed regions in cells with RNase H2 deficiency.⁴¹ In normal postmitotic neurons, ID4 indels are also strongly associated with transcription.³⁵ To explore the association between sIndels and transcription, we first split all sIndels into two categories: (1) all 2-bp deletions and the 2-bp microhomology deletions that characterize ID-A (Figure 3C) and (2) all other sIndels. We then compared the burdens of each sIndel category to local gene expression levels measured in excitatory neurons from previously published snRNA-seq data.³⁵ In neurotypical control neurons, both sIndel categories were strongly associated with expression levels (Figure 4F). In the neurodegenerative conditions (which contain higher amounts of ID-A-like sIndels), both categories remained positively associated with expression, though the association was weaker for ID-A-like sIndels. This association with transcriptional activity was confirmed by comparing each sIndel category against several other signals that correlate with transcriptional activity: local levels of open chromatin (measured by single-nucleus ATAC-seq³⁵), active and inactive histone marks,⁴⁶ chromatin states,⁴⁷ and promoters and enhancers active in neurons⁴⁸ (Figures S7B and S7C). Taken together, these characteristics of TOP1-related indel mutagenesis suggest that the ID-A-like sIndels observed in our diseased neurons are likely TOP1-mediated in these three neurodegenerative conditions.

Single-strand breaks and deletions are prevalent in neurodegenerative conditions

The removal of incorporated ribonucleotides during TOP1-mediated RER creates single-strand breaks (Figure 4A), and analysis of genomic DNA (gDNA) showed directly that cortex tissues from diseased cases with large numbers of sIndels exhibited widespread DNA fragmentation. To assess these single-strand breaks, gDNA extracted from disease-relevant regions of *C9ORF72* ALS, *C9ORF72* FTD, AD, and neurotypical control

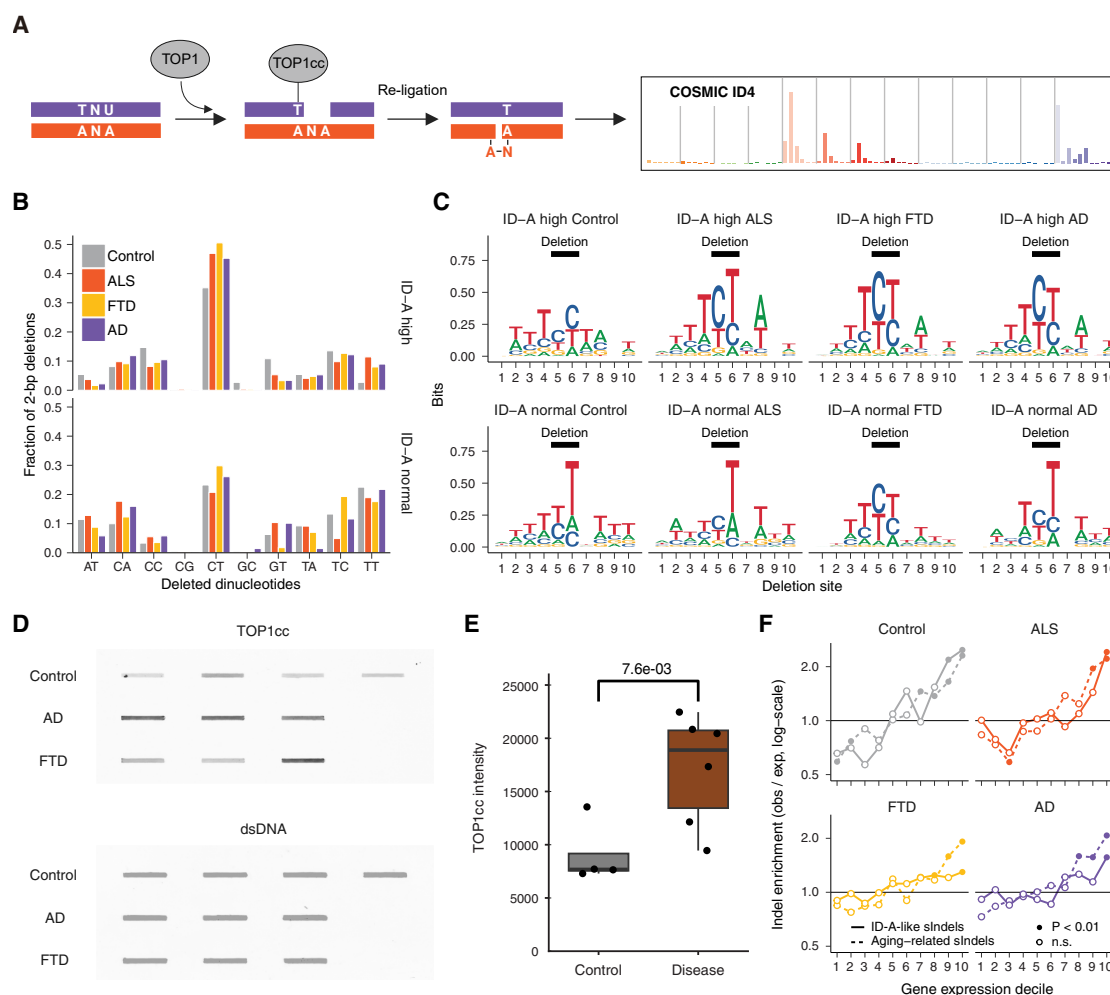


Figure 4. Disease-associated ID-A-like deletions reflect characteristics of TOP1-mediated RER

(A) Model of TOP1-mediated RER.

(B) Fraction of 2-bp deletions by the deleted dinucleotide stratified by ID-A-high and -normal classifications.

(C) DNA motifs showing the ten basepair context around 2-bp deletions with at least 2 repeat units and stratified by ID-A-high and -normal classifications. Positions 5 and 6 are the deleted nucleotides.

(D) RADAR assay measuring TOP1 cleavage complex (TOP1cc) levels in NeuN+ neuronal nuclei from *C9ORF72* FTD, AD, and control brains. Slot blots were probed with antibodies against TOP1cc and dsDNA. Each band represents one individual.

(E) Quantification of TOP1cc levels in arbitrary units in (D) using ImageJ. Each dot represents one individual. The *p* value represents an unpaired one-sided Welch's *t* test.

(F) Enrichment of indels in relation to local gene expression levels from previously published snRNA-seq data of excitatory neurons. Enrichment tests divide the genome into 10 roughly equally sized, non-contiguous regions ranked by gene expression. I.e., decile 10 represents the subset of the genome containing the top 10% of expressed genes. Solid lines: enrichment analysis of ID-A-like indels (see Figure 3C). Dotted lines: all other indels. *p* values are enrichment tests (see STAR Methods) indicating that the observed/expected ratio significantly differs from 1.

See also Figures S7 and S8

brains was treated with S1 nuclease and analyzed using agarose gel electrophoresis (Figure 5A; see STAR Methods). S1 nuclease converts single-strand breaks into double-strand breaks, which appear as smears on the gel. We then quantified smear intensity using TapeStation and found significantly elevated levels of fragmented single-strand gDNA in diseased brains (Figure 5B), with no correlation to donor age (Figure 5C). Furthermore, the levels of fragmented single-strand gDNA in diseased brains were significantly correlated with the burdens of ID-A indels, accounting for

62%, 77%, and 79% of ID-A variance in ALS, FTD, and AD, respectively (Figure 5D), strongly supporting that ID-A indels lead to DNA fragmentation in a disease-specific process. Interestingly, we also observed significantly increased levels of pre-existing double-strand breaks in diseased brains, which similarly correlated with ID-A burdens (Figures 5A–5D), suggesting that progressive accumulation of single-strand breaks may eventually lead to double-strand break formation and widespread genome instability in disease.

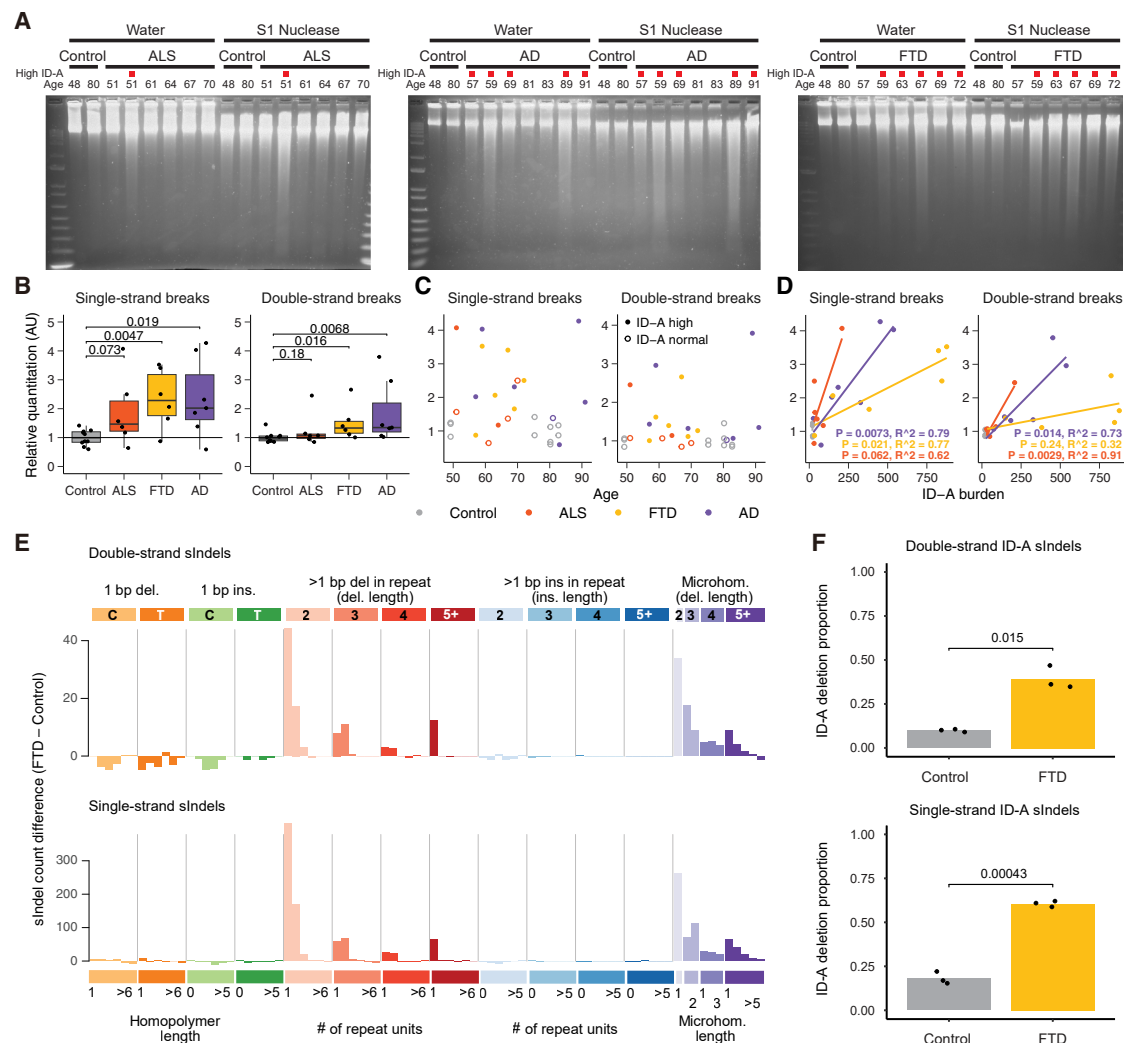


Figure 5. Double-strand ID-A sIndels associate with single-strand breaks and deletions

(A) Agarose gel electrophoresis of gDNA from motor and premotor cortex tissues of *C9ORF72* ALS and premotor cortex tissues of control brains, as well as from prefrontal cortex tissues of *C9ORF72* FTD, AD, and control brains. Samples are arranged in order of donor age. Increased smear intensity following S1 nuclease treatment indicates single-strand breaks. Red squares indicate ID-A-high cases determined from single-cell WGS (see STAR Methods) and strongly correlate with the appearance of increased smear intensity following S1 nuclease treatment on the gel.

(B) Mean single-strand break and double-strand break intensities for each condition (*C9ORF72* ALS, *C9ORF72* FTD, AD, and control). *p* values are Wilcoxon rank-sum tests. *y* axis values represent quantification of the smears in (A) in arbitrary units (a.u.) using Agilent TapeStation.

(C) Abundance of single-strand and double-strand break intensities from (B) plotted against the age of donor.

(D) Average ID-A burden per brain is correlated with single-strand break and double-strand break intensities quantified in (B) in *C9ORF72* ALS, *C9ORF72* FTD, and AD brains, suggesting a common process. Each point represents one brain. Averages are taken over all cortical neurons subjected to scWGS from each brain. *p* values are regression *t* tests on slope. *p* values for controls are not significant and omitted on the graph.

(E) Indel spectra derived from duplex sequencing of 50-neuron mini-bulk samples with META-CS. Spectra are shown by subtracting the mean spectrum of age-matched neurotypical controls from the mean spectrum of *C9ORF72* FTD cases. Spectra for double-strand and single-strand sIndels, which can be distinguished due to strand-specific tagging, are shown separately.

(F) FTD shows increased proportions of ID-A-like deletions among both single- and double-strand sIndels (numbers of ID-A-like deletions divided by total numbers of indels) compared with age-matched controls from META-CS duplex sequencing data. *p* values are two-tailed unpaired *t* tests.

Multiplexed end-tagging amplification of complementary strands (META-CS),⁴⁹ a type of duplex sequencing that distinguishes the Watson and Crick strands of DNA molecules, provided a third, orthogonal confirmation of the presence of single- and double-strand 2-bp deletions in diseased brains. During TOP1-mediated RER, the subsequent re-ligation of single-

strand breaks creates an intermediate DNA product containing a 2-bp single-strand deletion (Figure 4A). We performed META-CS on 50-neuron samples from the prefrontal cortex from three ID-A-high *C9ORF72* FTD brains and three age-matched control brains. The duplex sequencing data confirmed that ID-A-like, double-strand 2-bp deletions were enriched in

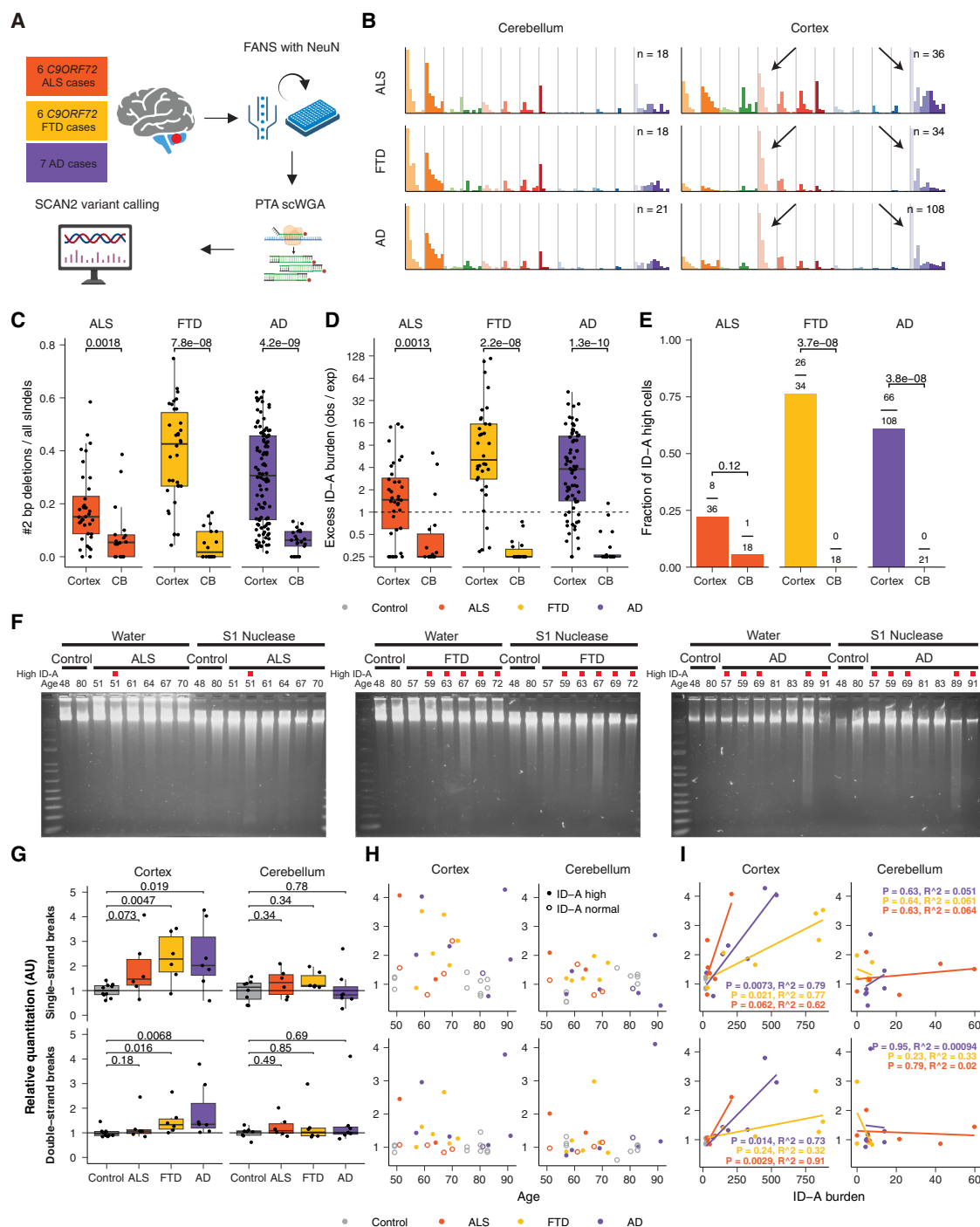


Figure 6. Cerebellar samples from *C9ORF72* ALS, *C9ORF72* FTD, and AD brains do not show evidence of TOP1-mediated mutagenesis

(A) Overview of the experimental design. Three cerebellar granule neuron nuclei were isolated from the cerebellum of each of the 6 *C9ORF72* ALS, 6 *C9ORF72* FTD, and 7 AD brains for which scWGS of cortical neurons were analyzed in this study. Cerebellar neurons were then amplified by PTA, subjected to genome sequencing, and analyzed using the same procedures as cortical neurons.

(B) Indel spectra of cerebellar neurons and cortical neurons from diseased brains.

(C) Fraction of 2-bp deletions in cortical and cerebellar neurons from diseased brains. p values represent Wilcoxon rank-sum tests. CB, cerebellum.

(D) Excess burdens of ID-A indels in cortical and cerebellar neurons from diseased brains are shown as observed/expected ratios. Expected values were derived from PTA-amplified control cortical neurons. p values represent Wilcoxon rank-sum tests.

(E) Fraction of cells with high levels of ID-A indels in cortical and cerebellar neurons from diseased brains, determined by a classifier (see STAR Methods). p values represent Fisher tests of independence between disease status and ID-A-high classifications.

(legend continued on next page)

C9ORF72 FTD brains, as evidenced by the same deletion being captured on the independently sequenced Watson and Crick strands. Duplex sequencing also captured a larger number of ID-A-like single-strand lesions in *C9ORF72* FTD brains, for which the deletion was absent on one of the two independently sequenced strands (Figure 5E). The proportions of ID-A-like deletions were significantly higher in *C9ORF72* FTD brains compared with control brains for both double- and single-strand events, but single-strand deletions showed a more prominent increase (Figure 5F).

These results suggest that the increased levels of TOP1-mediated RER in neurons from neurodegenerative conditions generate frequent single-strand breaks and deletions, a small fraction of which are misrepaired to create double-strand deletions. Consistent with this, diseased neurons show upregulated TOP1 expression but reduced expression of pathways responsible for clearing TOP1ccs (Figure S8). Together, these findings indicate that increased TOP1-mediated RER is a common mechanism underlying widespread, catastrophic DNA damage in neurodegenerative conditions.

TOP1-mediated mutagenesis is absent in the cerebellum of diseased brains

To determine whether the increased TOP1-induced DNA damage and somatic mutations observed in diseased neurons reflect region-specific pathological processes rather than nonspecific end-stage genomic DNA damage, we analyzed somatic mutations and DNA strand breaks in the cerebellum, a brain region that is relatively spared in ALS, FTD, and AD. We performed PTA-based scWGA and scWGS on three cerebellar granule cell neurons from each of the diseased brains for which cortical neurons were collected (Figures 6A and S9).^{50,51} Analysis of scWGS data revealed an absence of 2-bp deletions in the Sndel spectra of cerebellar neurons across all three neurodegenerative conditions (Figure 6B). Consistent with this, the fractions of 2-bp deletions were significantly lower in cerebellar neurons compared with cortical neurons (Figure 6C). The overall burdens of ID-A Sndels were also significantly lower in cerebellar neurons compared with cortical neurons (Figure 6D). Stratification of cerebellar neurons by ID-A status further demonstrated that only a single cerebellar neuron from a *C9ORF72* ALS brain met criteria for ID-A-high classification, whereas no ID-A-high cerebellar neurons were detected in *C9ORF72* FTD or AD cases (Figure 6E).

We next performed strand break analyses on gDNA extracted from cerebellum tissues of diseased and control brains. In contrast to cortical neurons, where diseased brains showed elevated strand break levels, levels of both single-strand breaks

and double-strand breaks in cerebellar gDNA were comparable between diseased and control brains (Figures 6F and 6G) and did not correlate with donor age or ID-A burdens in cerebellar neurons (Figures 6H and 6I). Together, these findings indicate that TOP1-mediated mutagenesis is largely absent in the cerebellum of diseased brains and is restricted to disease-relevant cortical regions, consistent with a region-specific contribution of TOP1-induced genomic damage to ALS, FTD, and AD pathogenesis.

Deficiency of RNase H2 and RNR in diseased neurons

During DNA repair, DNA polymerases must incorporate dNTPs and discriminate against rNTPs. This selectivity relies on polymerase properties, as well as the balance of dNTP and rNTP pools, which is regulated by ribonucleotide reductases (RNRs)⁵²: when the dNTP:rNTP ratio is low, rNTP incorporation increases, threatening genome stability. Embedded rNTPs are removed by RNase H2, but in its absence, TOP1 serves as an alternative repair pathway (Figure 7A). We found that genes encoding RNRs and RNase H2 subunits were downregulated across various neuronal subtypes in our snRNA-seq data from *C9ORF72* FTD brains compared with controls (Figure S10A). A similar trend of reduced expression was also observed in previously published snRNA-seq data from AD neurons (Figure S10A), although most of these changes were not statistically significant, likely due to the low abundance of these transcripts in neurons and differences in tissue lysis and nuclear isolation protocols. However, qPCR on cDNA from neurons of *C9ORF72* FTD and AD brains with high levels of ID-A Sndels and control brains confirmed significantly reduced expression of *RNASEH2B* and *RRM2B* in *C9ORF72* FTD and AD neurons compared with control neurons (Figures 7B and S10B).

Finally, we observed a strong correlation between ID-A and SBS-B—which resembles COSMIC SBS30, a signature associated with oxidative DNA damage—in cortical neurons from all three disease conditions after removing three outlier ALS neurons that exhibited elevated sSNV burdens, possibly driven by another mechanism not involving TOP1 (Figures 7C, S10C, and S11). The correlation between elevated sSNV and Sndels in the same neurons indicates that neurons that accumulate higher levels of DNA damage due to oxidized nucleotides are the same ones that show catastrophic increases in TOP1-mediated mutagenesis. This correlation could reflect the known tendency of DNA polymerases β and λ to erroneously incorporate ribonucleotides into the genome during excision repair of oxidative DNA damage.⁵³ The misincorporation of ribonucleotides under these conditions could exacerbate genomic instability, further amplifying TOP1-mediated mutagenesis (Figure 7A). Furthermore,

(F) Agarose gel electrophoresis of gDNA from cerebellum tissues of disease and control brains. Samples are arranged in order of donor age. Increased smear intensity following S1 nuclease treatment indicates single-strand breaks. Red squares indicate ID-A-high cases determined from single-cell WGS (see STAR Methods).

(G) Mean single-strand break and double-strand break intensities in cortex and cerebellum for each condition (*C9ORF72* ALS, *C9ORF72* FTD, AD, and control). *p* values are Wilcoxon rank-sum tests. *y* axis values represent quantification of the smears in (F) in arbitrary units (a.u.) using Agilent TapeStation.

(H) Abundance of single-strand and double-strand breaks in cortex and cerebellum vs. age.

(I) ID-A burden from single-cell WGS of cerebellar neurons (mean observed/expected ratio across neurons in each case) does not correlate with single-strand break and double-strand break intensities quantified in (F) for *C9ORF72* ALS, *C9ORF72* FTD, and AD brains. *p* values are regression *t* tests on slope. *p* values for controls are not significant and omitted on the graph.

See also Figure S9.

TOP1 is a DNA topoisomerase that facilitates transcription, DNA replication, and DNA repair by alleviating topological tensions in the DNA.^{55,56,58} During this process, TOP1 forms a transient cleavage complex (TOP1cc), in which it covalently binds to DNA and introduces a single-strand nick that is subsequently religated. In neurons, TOP1 has been implicated in the transcriptional regulation of long neuronal genes essential for synaptic function and plasticity,^{59,60} and depletion of TOP1 in neurons has been linked to early-onset neurodegeneration and autism.^{59,61} In the absence of RNase H2, TOP1 compensates by facilitating the removal of misincorporated ribonucleotides from the genome. Single-strand breaks induced by TOP1 can produce intermediate DNA products with single-strand 2-bp deletions, which were detected by our duplex sequencing approach (Figures 5E and 5F). Given the increased oxidative damage observed in neurons from the three neurodegenerative conditions manifested as increased oxidative sSNV signature exposure (Figures S4A and S4B; SBS-B), the increased level of single-strand breaks in *C9ORF72* FTD and AD brains is likely a result of TOP1cc trapping by local oxidative DNA damage (Figures 5A–5D). Thus, although sSNVs caused by oxidative stress may not be sufficiently numerous to affect the genome, the resulting catastrophic levels of somatic deletions present in some neurons could potentially contribute to neuronal dysfunction by impacting the transcription of hundreds or even thousands of genes. Furthermore, the strong correlation between ID-A deletions and single-strand breaks (Figures 5A–5D) suggests that DNA fragmentation—a hallmark feature of these neurodegenerative conditions^{9,11}—is not a result of poor tissue quality or other technical factors. Instead, it occurs during the neuron's lifetime as a consequence of TOP1 activity during RER and may be relevant to disease progression. In addition to oxidative DNA damage-induced TOP1cc trapping, impaired repair of trapped TOP1ccs may further contribute to their persistence. Our snRNA-seq analyses showed that genes involved in TOP1cc resolution are downregulated (Figure S8). These include components of the SPRTN metalloprotease-mediated proteolysis and testis-expressed protein 264 (TEX264)-dependent nucleophagy pathways,^{62,63} both of which are VCP/p97-dependent TOP1cc repair pathways. Importantly, VCP is a gene mutated in some familial ALS and FTD patients. Together, these observations suggest that impaired TOP1cc clearance may further exacerbate the accumulation of DNA damage and genomic fragmentation in vulnerable neurons.

The observation of a shared mutagenic mechanism between such categorically different conditions—TDP-43 (ALS and FTD) and tau proteinopathies (AD)—is both surprising and suggestive of even broader applicability to neurodegenerative disease and perhaps neuronal death in general. Indeed, a recent report identified ID-A-like deletions in chronic traumatic encephalopathy (CTE),⁶⁴ a second tauopathy caused by repetitive head trauma.⁶⁵ TDP-43 pathology has been observed in a subset of AD and CTE cases,^{66,67} raising the possibility that nuclear TDP-43 depletion is the underlying shared cause of ID-A deletions; however, this appears unlikely since we did not observe an association between ID-A deletions and TDP-43 depletion status in *C9ORF72* ALS and FTD brains (Figures 2D and S6), and TDP-43 knockdown in induced pluripotent stem cell (iPSC)-derived neurons did not

induce TOP1cc formation (Figure S12). Importantly, we also excluded the possibility that tissue preservation artifacts contribute to the observed mutational pattern. Brain samples were obtained from multiple independent sources (Table S1),⁶⁴ and levels of somatic mutations, particularly ID-A deletions, did not correlate with postmortem interval across samples (Figure S13). Together, these results indicate that neither TDP-43 depletion nor tissue quality accounts for the shared ID-A mutational signature. Instead, increased oxidative stress—implicated in all of these neurodegenerative disease conditions^{7,10,68}—may play a key role. The strong correlation between ID-A deletions and oxidative sSNV signature exposure (Figure 7C) supports oxidative stress as a plausible driver of ID-A deletion formation across a wide range of neurodegenerative conditions. Indeed, oxidative DNA damage can increase the erroneous incorporation of ribonucleotides into the genome during excision repair of oxidative DNA damage and elevate the burden of substrates requiring RNase H2-dependent RER. In addition, *RNASEH2A* expression is reduced in Werner syndrome, a progeroid disorder characterized by replication stress and DNA damage,⁶⁹ suggesting that genotoxic stress can suppress RNase H2 expression. Therefore, oxidative DNA damage may both increase ribonucleotide incorporation and secondarily reduce RNase H2 expression, thereby further fueling TOP1-dependent RER activity. One plausible upstream cause of the oxidative stress itself would be the inflammatory activation of microglial cells, which has been observed in all of these conditions⁷⁰ and which is associated with increased secretion of cytokines and reactive oxygen species that may damage nearby neurons.

The shared genomic pathology in diverse neurodegenerative conditions like ALS, FTD, and AD raises the question of how general this pattern of genomic damage is to other forms of neuronal degeneration. Since our ALS and FTD cohorts all harbored *C9ORF72* repeat expansions, our findings may not necessarily generalize to all ALS and FTD cases, especially those without known genetic mutations. Nonetheless, diseases such as ataxia telangiectasia and Aicardi-Goutières syndrome likely also involve ID-A deletions, as these conditions involve persistent TOP1cc or RNase H2-related RER deficiency,^{71,72} which our data suggest is predictive of ID-A deletions. Since our ALS and FTD cases involved *C9ORF72* repeat expansions, it is also reasonable to question whether other repeat expansion disorders would harbor ID-A deletions. Somatic expansion of unstable repeats, such as those in trinucleotide repeat disorders, often involves activation of various DNA damage repair pathways, which may deplete the cellular dNTP pools and exacerbate ribonucleotide incorporation into the genome.⁷³ Overall, our data support that preventing excessive or unresolved TOP1cc formation and/or improving RER activity—rather than direct TOP1 inhibition—may represent potential therapeutic targets in a broad spectrum of neurodegenerative conditions.

RESOURCE AVAILABILITY

Lead contact

Requests for further information and resources should be directed to and will be fulfilled by the lead contact, Christopher A. Walsh (christopher.walsh@childrens.harvard.edu).

Materials availability

This study did not generate any new, unique reagents.

Data and code availability

Newly generated bulk, single-cell and META-CS DNA sequencing and single-nucleus RNA sequencing data have been uploaded to dbGaP: phs003530.v2.p1. All newly generated sequencing data are available conditional on approval for controlled access, in line with privacy regulations. Previously published bulk whole-genome and single-cell DNA-, RNA-, and ATAC-sequencing data are available for download at dbGaP: phs001485.v3.p1, and NIAGADS: NG00162, subject to controlled data access approval. All scripts used to perform analyses are available on Zenodo: <https://doi.org/10.5281/zenodo.20174164>.

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AUTHOR CONTRIBUTIONS

Conceptualization, Z.Z., L.J.L., G.D., P.J.P., C.L.-T., E.A.L., and C.A.W.; methodology, Z.Z., L.J.L., G.D., D.D.S., W.J.N., and A.N.; resource, K.E., E.G., M.B.M., A.Y.H., and C.L.-T.; software, L.J.L. and G.D.; investigation, Z.Z., L.J.L., G.D., J. Kim, J. Ku, K.K., N.R., M.B., A.C., and B.S.; visualization, Z.Z., L.J.L., G.D., and J. Ku; funding acquisition, Z.Z., P.J.P., C.L.-T., E.A.L., and C.A.W.; project administration, Z.Z., L.J.L., P.J.P., C.L.-T., E.A.L., and C.A.W.; supervision, Z.Z., L.J.L., P.J.P., C.L.-T., E.A.L., and C.A.W.; writing – original draft, Z.Z., L.J.L., and G.D.; writing – review & editing, Z.Z., L.J.L., G.D., J. Kim, D.D.S., B.S., M.B.M., A.Y.H., P.J.P., C.L.-T., E.A.L., and C.A.W.

DECLARATION OF INTERESTS

P.J.P. is a member of the scientific advisory board for Bioskryb Genomics, Inc. C.L.-T. serves on the scientific advisory board and/or has received consulting fees from SOLA Biosciences, Libra Therapeutics, Arbor Biotechnologies, Dewpoint Therapeutics, Mitsubishi Tanabe Pharma Holdings America, Sanofi, AUTTX LLC, Carthera, the Milken Institute, and Applied Genetic Technologies Corporation. C.A.W. is a paid consultant (cash, no equity) to Third Rock Ventures and Flagship Pioneering (cash, no equity); is on the Clinical Advisory Board (cash and equity) of Maze Therapeutics; is a member of the scientific advisory board for Bioskryb Genomics, Inc.; and is a founding advisor for Mo-saica Medicines (equity). These companies did not fund or have any role in the conception or performance of this research project. The authors have an un-

published invention related to aspects of this work that may be subject to future patent protection.

STAR★METHODS

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SUPPLEMENTAL INFORMATION

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
Mouse monoclonal anti-NeuN Alexa Fluor 488 (Clone A60)	MilliporeSigma	Cat#MAB377X; RRID: AB_2149209
Mouse monoclonal anti-TARDBP Alexa Fluor 647 (clone 2E2-D3)	Abnova	Cat#H00023435-M01; RRID: AB_425904
Mouse monoclonal anti-Topoisomerase I-DNA Covalent Complexes (clone 1.1A)	MilliporeSigma	Cat#MABE1084; RRID: AB_2756354
Mouse monoclonal anti-ds DNA (clone 3519 DNA)	Abcam	Cat#ab27156; RRID: AB_470907
Rabbit polyclonal anti-TDP-43 (N-terminal)	Proteintech	Cat#10782-2-AP; RRID: AB_615042
Chicken polyclonal anti-MAP2 (amino acids 235–1588)	Thermo Fisher	Cat#PA1-10005; RRID: AB_1076848
Donkey anti-rabbit IgG (H+L) Alexa Fluor 488	Thermo Fisher	Cat#A-21206; RRID: AB_2535792
Donkey anti-Chicken IgY (H+L) Alexa Fluor 647	Thermo Fisher	Cat#A78952; RRID: AB_2921074
Biological samples		
Brain tissue samples from ALS, FTD, AD and control cases	NIH NeuroBioBank; Massachusetts Alzheimer's Disease Research Center Brain Bank; Boston University UNITE Brain Bank	see Table S1
Chemicals, peptides, and recombinant proteins		
DAPI, FluoroPure™ grade	Thermo Fisher Scientific	D21490
AMPure XP beads	Beckman Coulter	A63882
SPRIselect beads	Beckman Coulter	B23317
Halt™ Protease Inhibitor Cocktail	Thermo Fisher	87785
DNAzol	Thermo Fisher	10503027
N-lauroylsarcosine sodium salt	MilliporeSigma	L7414
Glycogen	Thermo Fisher	R0561
Intercept® (TBS) Blocking Buffer	LI-COR	927-60001
Thermolabile Proteinase K	New England Biolabs	P8111S
Tn5 transposase	Diagenode	C01070010
Tagmentase Dilution Buffer	Diagenode	C01070011
Q5 High-Fidelity DNA Polymerase	New England Biolabs	M0491
Thermolabile Exonuclease I	New England Biolabs	M058
S1 nuclease	Thermo Fisher	EN0321
SYBR™ Gold Nucleic Acid Gel Stain	Thermo Fisher	S11494
Critical commercial assays		
Chromium Next GEM Single Cell 3' GEM, Library & Gel Bead Kit v3.1, 16 rxns	10X Genomics	PN-1000121
Chromium Next GEM Automated Chip G Single Cell Kit, 48 rxns	10X Genomics	PN-1000136
Dual Index Kit TT Set	10X Genomics	PN-1000215
ResolveDNA Whole Genome Amplification Kit	BioSkryb Genomics	100136
KAPA HyperPlus Kit	Roche	07962428001
KAPA UDI Adapters Kit 15 μM	Roche	08861919702

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
Phusion Hot Start II DNA Polymerase Kit	Thermo Fisher Scientific	F549L
EZ1 DNA Tissue Kit	Qiagen	953034
Quant-iT™ dsDNA Assay Kits, high sensitivity	Thermo Fisher Scientific	Q33120
Qubit™ 1X dsDNA High Sensitivity Assay Kit	Thermo Fisher	Q33231
High Sensitivity D5000 Reagents	Agilent Technologies	5067–5593
High Sensitivity D5000 ScreenTape	Agilent Technologies	5067–5592
Genomic DNA ScreenTape	Agilent Technologies	5067–5365
Genomic DNA Reagents	Agilent Technologies	5067–5366
High Sensitivity D1000 ScreenTape	Agilent Technologies	5067–5584
High Sensitivity D1000 Reagents	Agilent Technologies	5067–5585
NEBNext® Multiplex Oligos for Illumina	New England Biolabs	E7335S, E7500S, E7710S, E7730S
Power SYBR™ Green Cells-to-CT™ Kit	Thermo Fisher	4402954

Deposited data

Aligned single-cell and bulk DNA sequencing data for ALS and FTD cortical neurons	This study	dbGaP: phs003530.v2.p1
Single-nucleus RNA sequencing of ALS, FTD and control brain tissues	This study	dbGaP: phs003530.v2.p1
Aligned single-cell DNA sequencing data for ALS, FTD and AD cerebellar neurons	This study	dbGaP: phs003530.v2.p1
Single neuron (PTA) and matched bulk whole genome sequencing data	Luquette et al. ²⁶	dbGaP: phs001485.v3.p1
Single-nucleus RNA and ATAC sequencing of control brains	Ganz et al. ³⁵	dbGaP: phs001485.v3.p1; NIAGADS: NG00162
Single oligodendrocyte whole genome sequencing data	Ganz et al. ³⁵	NIAGADS: NG00162

Experimental models: Cell lines

KOLF2.1J	The Jackson Laboratory (iPSC Neurodegenerative Disease Initiative (INDI))	JIPSC001000
FA11	NINDS	NDS00242

Oligonucleotides

Human RNASEH2A primers for qPCR Forward: 5'-ACAGCCACTGGGCTTATACAG -3' Reverse: 5'-TCCCTACGGTGTCCACGAATA -3'	PrimerBank	https://pga.mgh.harvard.edu/primerbank
Human RNASEH2B primers for qPCR Forward: 5'-TAACCCCTGTTTCAGGAGAAGG -3' Reverse: 5'-ACACGTTATCCACCACAATTG -3'	PrimerBank	https://pga.mgh.harvard.edu/primerbank
Human RNASEH2C primers for qPCR Forward: 5'-AGGGACTCGAAGTGTCGTTTC -3' Reverse: 5'-TGTCACCATCACGTATCCAC -3'	PrimerBank	https://pga.mgh.harvard.edu/primerbank
Human RRM1 primers for qPCR Forward: 5'-GCCAGGATCGCTGTCTCTAAC -3' Reverse: 5'-GAGAGTGTTGCCATTATGTGGA -3'	PrimerBank	https://pga.mgh.harvard.edu/primerbank

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Continued

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Human RRM2B primers for qPCR Forward: 5'- ATTGGGCCTTGCGA TGGATAG -3' Reverse: 5'- GAGTCCTGGCATA AGACCTCT -3'	PrimerBank	https://pga.mgh.harvard.edu/primerbank
Human RBFOX3 primers for qPCR Forward: 5'- TCGTAGAGGGACGG AAAATTGA -3' Reverse: 5'- GCCGTTGGTGT AGGGGTTC -3'	PrimerBank	https://pga.mgh.harvard.edu/primerbank
Recombinant DNA		
PB-ZIM3-dCas9-BFP	Laboratory of Michael Ward	N/A
CROPseq-multi-Puro (pRW1441) dual-sgRNA vector	Walton et al. ⁷⁴	Addgene Plasmid # 216217
PB-TO-hNGN2-BSD-mApple	iPSC Neurodegenerative Disease Initiative (iNDI) & Laboratory of Michael Ward	Addgene Plasmid # 182308
Software and algorithms		
bwa mem 0.7.15-r1140	Li ⁸¹	https://github.com/lh3/bwa
GATK 3.5.0-g36282e4	Broad Institute	https://gatk.broadinstitute.org
SCAN2 1.2.13	Luquette et al. ²⁶	https://github.com/parklab/SCAN2
SigProfilerMatrixGenerator 1.2.26	Bergstrom et al. ⁷⁶	https://github.com/AlexandrovLab/SigProfilerMatrixGenerator
SigProfilerExtractor 1.1.24	Islam et al. ⁷⁷	https://github.com/AlexandrovLab/SigProfilerExtractor
Umap 0.2.10.0 (R package)	CRAN	https://github.com/tkonopka/umap
Cellranger 6.1.0	10X Genomics	https://www.10xgenomics.com/support/software/cell-ranger/latest
CellBender 0.3.0	Broad Institute	https://github.com/broadinstitute/CellBender
scDblFinder 1.16.0	Germain et al. ⁸⁵	https://github.com/plger/scDblFinder
DropletQC 1.0.0	Muskovic et al. ⁸⁶	https://github.com/powellgenomicslab/DropletQC
Seurat 5.1.0	Hao et al. ⁸⁷	https://satijalab.org/seurat/
Harmony 1.2.0 (R package)	CRAN	https://github.com/immunogenomics/harmony
Rstan 2.32.6 (R package)	CRAN	https://github.com/cran/rstan/releases
bayestestR 0.13.2 (R package)	CRAN	https://github.com/easystats/bayestestR
Nebula 1.5.3 (R package)	CRAN	https://github.com/lhe17/nebula
Regtools 1.7.0 (R package)	CRAN	https://github.com/griffithlab/regtools
Duplex-Indel 1.0.1	Dong et al. ⁸²	https://github.com/ealee-lab/duplex-indel
Custom scripts for figures and analysis	This study	Zenodo: https://doi.org/10.5281/zenodo.20174164

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Human brain tissue sources

Fresh frozen postmortem human brain tissues were originally collected by the Massachusetts Alzheimer's Disease Research Center and NIH NeuroBioBank (Table S1) according to their respective institutional protocols, written authorization and informed consent; they were subsequently obtained for this study with the approval of the Boston Children's Hospital Institutional Review Board. Research on these deidentified specimens and data was performed at Boston Children's Hospital with approval from the Committee on Clinical Investigation. The presence of C9ORF72 repeat expansion in these cases were determined through repeat-primed PCR (RP-PCR).

METHOD DETAILS

Neuronal nuclei isolation

Single neuronal nuclei were isolated using FANS with co-staining of NeuN and TDP-43 antibodies.^{27,75} Fresh frozen postmortem human brain tissues were homogenized in a Dounce homogenizer with a chilled tissue lysis buffer (10 mM Tris-HCl, 0.32 M sucrose, 3 mM Mg(OAc)₂, 5 mM CaCl₂, 0.1 mM EDTA, 1 mM DTT, 0.1% Triton X-100, pH 8) on ice. Tissue lysates were carefully overlaid on top of a sucrose cushion buffer (1.8 M sucrose, 3 mM Mg(OAc)₂, 10 mM Tris-HCl, 1 mM DTT, pH 8) and ultra-centrifuged for 1 hour at 30,000 x g. Nuclear pellets were incubated and resuspended in ice-cold PBS supplemented with 3 mM MgCl₂, filtered (40 μm pore size), then stained with Alexa Fluor 488-conjugated anti-NeuN antibody (Millipore MAB377X) and Alexa Fluor 647-conjugated anti-TDP-43 antibody (Abnova H00023435-M01). DAPI was added to stained nuclei right before the sorting. Single neuronal nuclei with TDP-43 pathology (NeuN+, TDP-43-) and without TDP-43 pathology (NeuN+, TDP-43+) were then sorted into individual wells of 96-well plates

Single-cell whole-genome amplification

Primary template-directed amplification (PTA) of sorted single nuclei was performed using the ResolveDNA® Whole Genome Amplification Kit v1 (BioSkrbyb 100136) according to the manufacturer's protocol.²⁶ Amplified single-cell genomes were subjected to quality control using Picogreen quantification and a 4-locus multiplex PCR assay to assess DNA yield and amplification evenness.⁷⁵ Single-cell genomes with sufficient DNA yield and successful amplification of all four loci were selected for single-cell whole-genome sequencing (scWGS).

Library preparation for scWGS

Libraries were made from PTA amplified single-cell genomes following a modified KAPA HyperPlus Library Preparation protocol provided by BioSkrbyb.²⁶ Libraries were subjected to quality control using Picogreen quantification and TapeStation HS D1000 Screen Tape (Agilent PN 5067–5584) before sequencing. Single cell genome libraries were sequenced on the Illumina NovaSeq platform (150 bp x 2) at 30X for cortical neurons and 22X for cerebellar neurons.

scWGS data preprocessing

Illumina reads were aligned to the human reference with decoy sequence GRCh37d5 (hs37d5) using bwa mem version 0.7.15-r1140 with the -M option. Following alignment, BAMs were processed according to the GATK Best Practices recommendations, namely Picard MarkDuplicates version 2.8.0 and IndelRealigner, BaseRecalibrator and PrintReads using GATK version 3.5.0-g36282e4.

scWGS somatic mutation calling with SCAN2

Somatic SNVs and Indels were called from PTA and MDA data using SCAN2 version 1.2.13, git commit 09ed5b9 and r-scan2 git commit c0aef34. The standard SCAN2 recommended workflow was followed. First, to enable somatic Indel calling, scan2 makepanel was run on 8 separate batches of PTA and MDA single cells and matched bulks to make the analysis tractable. The samples included in each batch are defined in the scripts provided at <https://doi.org/10.5281/zenodo.20174164>. Each batch included the 56 control PTA cortical neurons and matched bulks generated in Ganz et al.³⁵ to identify recurrent indel artifacts with at least the same specificity as previous published analyses. Second, one instance of scan2 call_mutations was run for each unique individual using the makepanel output corresponding to that individual's batch. For each run, all BAMs belonging to the individual (i.e., all MDA, PTA and matched bulk(s)) were provided to SCAN2. Third, one instance of SCAN2's mutation-signature based rescue (scan2 rescue) was run for each unique combination of amplification type (PTA or MDA), phenotype (ALS, FTD, AD or control) and cell type (neuron, oligodendrocyte or mixed glia). Finally, SCAN2's post-processing script to filter clustered or recurrent (i.e., called in >1 single cell) artifacts and mutations (SCAN2/bin/digest_calls.R) was applied to each scan2 rescue output. For mutation enrichment analysis, only calls with final.filter=FALSE were retained. Critically, because of the unique mutational signature of disease-associated Indels, we opted to exclude all SCAN2 rescue mutation calls from all analyses to avoid any bias involved in signature-based rescue. Rescued calls were removed by excluding rows with rescue=TRUE from the final recurrence filtered output. Additional configuration options applicable to all SCAN2 steps were -gatk=sentieon_joint; the GRCh37 human reference genome with decoy hs37d5 (-ref), dbSNP v147 common (-dbSNP) and 1000 Genomes phase 3 SHAPEIT2 phasing panel (-shapeit-refpanel).

Single-cell QC and SCAN2 performance metrics

Technical metrics (mean sequencing depth, MAPD) and mutation detection metrics (sensitivity, fraction of genome analyzable and false discovery rates) were obtained using SCAN2's callstats() function in R. The median absolute pairwise difference (MAPD) values were extracted from each single cell using SCAN2's mapd() function. MAPD measures non-uniformity of amplification by examining the difference in read count between neighboring genomic bins: $MAPD = \text{median}_i \{ |\log_2 R_i - \log_2 R_{i+1}| \}$ where R_i is the GC-corrected read depth in genomic bin i divided by the average GC-corrected read depth across the genome. The set of genomic bins is distributed with the SCAN2 package and aims to create bins with an equal number of alignable bases using paired end 150 bp Illumina reads. MAPDs reported here used a bin size of 64 kb. SCAN2 estimates somatic SNV and Indel detection sensitivity by applying somatic criteria to known germline variants detected in the matched bulk.²⁶ The fraction of genome analyzable for each single cell is the

number of positions in the reference genome hs37d5 that meet the minimum sequencing read depth in the matched bulk (>10 reads) and single cell (>5 reads for sSNVs, >9 reads for sIndels) divided by the number of positions accessible by short read sequencing, estimated by the number of positions with >0 reads in the matched bulk. False discovery rates (FDRs) were derived from previous estimates of false positive errors committed by SCAN2 per megabase of genome analyzed that relied on *in silico* benchmarking experiments that preserved single-cell artifacts.²⁶ Briefly, read data from haploid X chromosomes from two male single cells were combined to create a synthetic diploid X chromosome, which was then analyzed by SCAN2. True somatic mutations—which should have allele fractions of ~100% on haploid chromosomes—were identified prior to combining the read data. Any SCAN2 mutation call not identified as a true mutation was counted as a false positive. Pseudoautosomal regions were excluded from this analysis. This process was performed for several pairs of male single cells to arrive at a false positive rate of 0.013 sSNV errors per Mb and 0.00073 sIndel errors per Mb. For each single cell and mutation type (SNV or indel), the false positive load is the product of multiplying the respective rate by the respective number of analyzable bases in that cell. The per-cell FDR is the false positive load divided by the total number of mutation calls or 1, whichever is smaller. The catalog-wide FDR is the sum of false positive loads over all cells divided by the sum of calls over all cells.

scWGS mutation burden estimation and comparison

SCAN2 estimates the genome-wide mutation burden by correcting the number of called mutations for detection sensitivity. This correction accounts for sensitivity changes caused by sequencing depth, imbalanced and/or non-uniform single-cell amplification, and other factors. Correction is performed separately for sSNVs and sIndels. More details are provided in our previous publication.²⁶ SCAN2's sSNV and sIndel mutation burden estimates were extracted from each SCAN2 object in R using `r-scan2's mutburden()` function. To compare mutational burdens between any two groups (e.g., control and ALS), it is first necessary to correct for differences in age between the two groups. Two strategies were used together to achieve this goal. First, we computed the expected burden as a function of age using PTA-amplified, neurotypical control neurons. This process involved two steps of linear models; the first to identify strong outliers and the second to compute the final age-based expectation. In more detail, the first linear model was fit to all PTA control neurons via $\text{lm}(\text{burden} \sim \text{age})$ and Tukey's method was applied to the model's residuals to identify strong outliers. The second linear model was then fit with the outliers removed and R's predict function was used to derive the expected burden as a function of age. Separate models were fit to sSNV, sIndel and mutational signature burdens (see *Mutational signature analysis* for more details on signature burdens); for analysis of oligodendrocytes, $n = 66$ PTA-amplified control oligodendrocytes were used in place of PTA control neurons. For each cell and burden type (i.e., sSNV, sIndel, etc.) a final observed/expected ratio was produced by dividing the observed burden from SCAN2 by the expected burden based only on the cell's age from the linear model. Second, for all burden comparisons between groups, only observed/expected ratios from cells with age > 50 were included since the residuals of younger cells tended to be large compared to the trend line, leading to inflated observed/expected ratios. Significance tests for differences in burdens between groups are Wilcoxon rank-sum tests on the observed/expected ratios. After acceptance of the manuscript, we identified that control case UMB5451 had been incorrectly annotated as female instead of male. Reanalysis using the corrected sex annotation showed no meaningful changes in mutation calls or reported results. Because SCAN2 performs mutation calling independently of sample sex and the analyses presented in this study report only autosomal mutation burdens, the incorrect sex annotation did not materially affect the conclusions of the study. Therefore, the figures and tables in the manuscript were not modified.

scWGS quality control and removal of outliers

To ensure that technical factors did not drive differences in burdens or mutational signatures, we compared sSNV and sIndel burdens against a quantity that reflects quality of single cell amplification, the median absolute pairwise difference (MAPD).³⁶ Low MAPD values indicate more uniform amplification and thus better quality. Plots of mutation burden vs. MAPD were visually assessed to identify cells in which technical quality may have driven high sIndel burdens (i.e., cells with both high sIndel and high MAPD). This revealed 5 PTA-amplified neurons (3 AD, 2 FTD) and 7 MDA-amplified neurons (6 AD, 1 control), which were excluded from further analysis. Additionally, 10 MDA-amplified neurons were excluded due to extreme sIndel burden and/or a highly atypical mutation signature enriched in long microhomology deletions.

Mutational signature analysis

Matrices of SBS96 and ID83 somatic mutation counts were generated with SigProfilerMatrixGenerator version 1.2.26 using the GRCh37 human reference genome.⁷⁶ All single cells (both MDA- and PTA-amplified; [Table S2](#)), including outliers, were included for matrix generation. Next, sIndel count matrices were corrected to reflect different sensitivities for the 83 Indel channels, as estimated in our previous publication.²⁶ De novo mutational signatures were then extracted using SigProfilerExtractor version 1.1.24 and options `min_stability=0.9`, `minimum_signatures=1`, `maximum_signatures=8`, and `nmf_replicates=100`.⁷⁷ The decompositions with the most stable signatures were selected, which was $K=6$ signatures for SBS96 and $K=3$ signatures for ID83. For each de novo signature, a genome-wide burden was computed by multiplying the SigProfilerExtractor assigned number of mutations by SCAN2's genome-wide scaling factor for the appropriate mutation type (i.e., SBS96 counts were multiplied by SCAN2's sSNV genome-wide burden scaling factor and ID83 counts were multiplied by SCAN2's sIndel genome-wide scaling factor). These scaled

genome-wide burdens were then used to fit models on PTA-amplified control neurons and calculate age-adjusted expected values using the procedure detailed in *scWGS mutation burden estimation and comparison*.

Classification of ID-A high and normal cells

To label each single cell either ID-A normal (typical levels of ID-A) or ID-A high (excessive ID-A, consistent with disease progression), we built a classifier based on 83-dimensional Sndel spectra. A matrix of Sndel spectra was first constructed using the same cells and corrected indel spectra described in *Mutational signature analysis*. We then used the R umap package to perform dimensional reduction from the 83-dimensional spectra into 2-dimensional space via the function call `umap(M, preserve.seed=FALSE, random_state=1, transform_state=1)`, where M represents the matrix of indel spectra. Under this dimensional reduction, all cells visually clustered into two distinct populations, one of which corresponded to high levels of ID-A. Cells in the cluster with elevated ID-A levels were classified as ID-A high; all other cells were classified as ID-A normal. Cells with no Sndel calls ($n = 2$ PTA-amplified control cortical neurons and $n = 3$ PTA-amplified control oligodendrocytes) were not classified.

Mutation enrichment analysis

Enrichment tests

Enrichment analysis was performed following the procedures outlined in our previous publications.^{26,35} Briefly, enrichment analysis considers the number of somatic mutations over a set of genomic regions which are not, in general, contiguous (e.g., defined by co-variables such as gene expression). The null hypothesis is that somatic mutations are uniformly distributed over the genome. To test the null hypothesis, SCAN2 simulates uniformly distributed mutations across the genome for each single cell, considering both (1) the regions of the genome with sufficient read depth to detect somatic mutations and (2) the mutational spectrum of the whole mutation set, which is required to match the observed somatic mutations. This simulation is performed 10,000 times to produce 10,000 null sets of mutations. The true mutation sets and null sets of mutations are then combined for each set of single cells of interest (e.g., all PTA single neurons from ALS). For each genomic region, the true mutation set and simulation sets are intersected with the region and tallied. The enrichment over a region is defined as the number of true mutations in the region divided by the mean number of simulated mutations in the region over the 10,000 simulations; values > 1 indicate enrichment while values < 1 indicate depletion. This same metric is applied to all 10,000 simulations to construct a distribution of the enrichment statistic. A two-sided enrichment test P -value is constructed by counting the number of simulations with more extreme enrichment ratios than the observed mutation count (to make this test two-sided, more extreme is defined as having greater absolute log-enrichment).

Enrichment regions

Seven diverse region definitions were used to assess mutation enrichment. (1) The ChromHMM 15-state model from dorsolateral prefrontal cortex (ID: E073), which assigns every genomic location one of 15 states⁴⁶; (2) transcribed and untranscribed genomic regions as determined by the GENCODEv26 gene model (UTRs, exons and introns were defined as transcribed); (3) promoters and enhancers active in neurons⁴⁸; (4) ChIP-seq measurements of histone modifications H3K27ac, H3K4me1, H3K4me3, H3K9ac, H3K9me3 from the Roadmap Epigenomics Project⁴⁶; (5) RNA-sequencing gene expression levels in the prefrontal cortex (BA9) from the GTEx Portal, v8 release⁷⁸; (6) single nucleus RNA-sequencing gene expression levels from excitatory neurons³⁵; and (7) single nucleus ATAC-sequencing of accessible chromatin in excitatory neurons.³⁵

Single-nucleus RNA sequencing library preparation

For each of the premotor cortex samples of C9ORF72 ALS brains and prefrontal cortex samples of C9ORF72 FTD brains, ten thousand neuronal nuclei with TDP-43 pathology (NeuN+, TDP-43-) and without TDP-43 pathology (NeuN+, TDP-43+) were sorted into individual wells of 96-well plates. For each of the premotor cortex samples and prefrontal cortex samples of age-matched normal brains, only ten thousand neuronal nuclei without TDP-43 pathology (NeuN+, TDP-43+) were sorted. Sorted nuclei were then used for droplet generation and sequencing library preparation using the 10x Genomics Next GEM Single Cell 3' GEM Kit v3.1 and Chromium Controller, following the manufacturer's manual.

Cryptic exon junction analysis

Annotated TDP-43 dependent cryptic junctions were obtained from the previous study and converted from hg38 to hg19 genome assembly.²⁸ Cryptic exon junction analysis was performed following the recent study with some modifications.²⁹ BAM files generated by cellranger were filtered to exclusively include reads meeting the criteria set in the snRNA-seq analysis (described above). For neuron type-specific junction analysis, BAM files were split into distinct neuron type-specific BAM files based on each cell's 10X barcode and corresponding cluster annotation. These filtered BAM files were subsequently analyzed using regtools' junctions extract function with the parameters “-a 6 -m 30 -M 500000 -s RF” to generate junction files.⁷⁹ Junction files were processed using Leafcutter's leafcutter_cluster_regtools.py with the parameters “-m 10 -p 0.0001” to generate a summary matrix of junction counts across the dataset.⁸⁰ Given the limited number of reads supporting individual splice junctions, read counts from two replicates were summed for downstream analysis. Only genes with at least 3 reads supporting the splicing events were included (defined as the sum of constitutive splicing reads and cryptic exonization reads; corresponding to the denominator in the regtools output).

TOP1cc flow cytometry assay

Nuclei were isolated from diseased and control brain tissues as described above. Purified nuclei were fixed using 2% paraformaldehyde (PFA) at 4 °C for 15 min, followed by a wash with ice-cold cell buffer (PBS supplemented with 3 mM MgCl₂). The nuclei were then stained with Alexa Fluor 488-conjugated anti-NeuN antibody (Millipore MAB377X) and Alexa Fluor 647-conjugated anti-TOP1cc antibody (Millipore MABE1084) in cell buffer containing 0.1% Triton-X100 and 1% bovine serum albumin (BSA) at 4 °C for 1 hr. DAPI was added to stained nuclei immediately before flow cytometry analysis. NeuN+/TOP1cc+ nuclei were analyzed using a BD FACSARIA II.

TOP1cc RADAR assay

Nuclei from diseased and control brain tissues were isolated as described above, with Halt™ Protease Inhibitor Cocktail (Thermo Fisher 87785) included in all buffers. For each case, 5 × 10⁵ nuclei were sorted into 300 μL of ice-cold PBS supplemented with 3 mM MgCl₂ in 1.5-mL DNA LoBind® tubes (Eppendorf 022431021). Sorted nuclei were centrifuged at 900x g for 10 min at 4 °C, and the supernatant was carefully removed without disturbing the nuclear pellet.

The pellet was lysed by adding 1 mL of DNAzol (Thermo Fisher 10503027) containing 1% N-lauroylsarcosine sodium salt (MilliporeSigma L7414), freshly prepared at room temperature. The nuclei were mixed by pipetting 10 times and maintained on ice for all subsequent steps. Glycogen (20 μg; Thermo Fisher R0561) was added to each sample, followed by thorough mixing. Each sample was then equally divided into two 1.5-mL DNA LoBind® tubes.

DNA was precipitated by adding 1 mL of ice-cold 100% ethanol and incubating at -20 °C for 10 min, followed by centrifugation at maximum speed for 10 min at 4 °C. The supernatant was carefully removed, and the DNA pellet was washed twice with ice-cold 80% ethanol, with centrifugation at maximum speed for 10 min at 4 °C after each wash. Pellets were air-dried for 3 min at room temperature and resuspended in 100 μL of 8 mM NaOH. Immediately after resuspension, 1 μL of 1 M HEPES buffer was added to neutralize the solution.

DNA concentrations were measured using the Qubit™ 1X dsDNA High Sensitivity Assay Kit (Thermo Fisher Q33231). For slot blot analysis, 300 ng of DNA from each sample was applied to a nitrocellulose membrane (Bio-Rad 1620115) using the Bio-Dot® SF Microfiltration Apparatus (Bio-Rad 1706542) according to the manufacturer's instructions. Membranes were UV-crosslinked for 10 min and blocked in Intercept® (TBS) Blocking Buffer (LI-COR 927–60001) for 1 h at room temperature with gentle rocking. Membranes were incubated with anti-TOP1cc antibody (Millipore MABE1084) or anti-ds DNA antibody (Abcam ab27156) diluted in Intercept® (TBS) Blocking Buffer for 2 h at room temperature. After three washes in TBST (10 min each), membranes were incubated with LI-COR IRDye® secondary antibodies and imaged using the LI-COR Odyssey Imaging System (model 9120). Signal intensities were quantified using ImageJ.

META-CS library preparation

META-CS libraries were made following the previously described protocol with modifications (DOI: <https://doi.org/10.17504/protocols.io.6qpvr3nbzvmk/v1>).⁴⁹ Fifty neuronal nuclei were sorted into individual wells of 96-well plates containing 2 μL of META lysis buffer (20 mM Tris, pH 8.0, 20 mM NaCl, 0.15% Triton X-100, 25 mM dithiothreitol, 1 mM EDTA, 3 units/mL NEB Thermolabile Proteinase K [P8111S]) at 30 °C for 1 h, 55 °C for 10 min. Lysed cells were stored at -80 °C until library preparation.

The transposome was assembled using Tn5 transposase (Diagenode C01070010) following manufacturer's manual (5 μL of annealed META mix, 5 μL of Tn5, 5 μL of glycerol). The assembled transposome was further diluted using the Tagmentase Dilution Buffer (Diagenode C01070011) based on optimized Tn5 concentration (performed on the day of library preparation). After thawing on ice, cell lysate was transposed with 8 μL of transposition mix (1 μL of diluted transposome, 5 μL of 2X Tagmentation Buffer [Diagenode C01019043], 2 μL of water) and incubated at 55 °C for 15 min. The transposition was stopped by adding 2 μL of 6X stop buffer (300 mM NaCl, 45 mM EDTA, 0.01% Triton X-100, 3 units/mL Thermolabile Proteinase K [NEB P8111S]) and incubated at 37 °C for 30 min, 55 °C for 10 min.

First-strand tagging was performed by adding 13 μL of Strand Tagging Mix 1 (5 μL of Q5 reaction buffer, 5 μL of Q5 high GC enhancer, 0.85 μL of 100 μM [total] Adp1 primer mix, 0.6 μL of 100 mM MgCl₂, 0.6 μL of water, 0.5 μL of 10 mM dNTP mix, 0.25 μL of 20 mg/mL bovine serum albumin [NEB B9200], 0.25 μL of Q5 DNA polymerase [NEB M0491]) and incubated at 72 °C for 3 min, 98 °C for 30 s, 62 °C for 5 min, 72 °C for 1 min. Adp1 primers were removed by adding 1 μL of Thermolabile Exol (NEB M058) and incubated at 37 °C for 15 min, 65 °C for 5 min.

Second-strand tagging was performed by adding 4 μL of Strand Tagging Mix 2 (1 μL of Q5 reaction buffer, 1 μL of Q5 high GC enhancer, 0.95 μL of 100 μM [total] Adp2 primer mix, 0.945 μL of water, 0.1 μL of 10 mM dNTP mix, 0.05 μL of Q5 DNA polymerase) and incubated at 98 °C for 30 s, 62 °C for 5 min, 72 °C for 1 min. Adp2 primers were removed by adding 1 μL of Thermolabile Exol (NEB M058) and incubated at 37 °C for 15 min, 65 °C for 5 min.

Strand tagging products were amplified by adding 19 μL of PCR mix (5 μL of NEBNext Multiplex Oligos Universal Primer, 5 μL of NEB Index Primers [NEB E7335S, E7500S, E7710S, E7730S], 4 μL of Q5 reaction buffer, 4 μL of Q5 high GC enhancer, 0.4 μL of 10 mM dNTP mix, 0.4 μL of water, 0.2 μL of Q5 DNA polymerase) and incubated at 98 °C for 20 s, 11 cycles of (98 °C for 10 s, 72 °C for 2 min), 72 °C for 2 min.

Size selection of META-CS libraries were performed using AMPure XP beads (Beckman Coulter A63882) with a 0.65X upper cut and a 1.8X lower cut following manufacturer's manual. Libraries were eluted into 30 μ L of low TE buffer. Each META-CS library was sequenced on one lane of the Illumina HiSeq X sequencer.

Strand break assay

gDNA was extracted from disease-relevant brain regions and cerebellum of *C9ORF72* ALS, *C9ORF72* FTD, AD, and control brains using the Qiagen EZ1 Advanced XL system (Qiagen 9001874) in a single batch. For each sample, two parallel reactions were prepared: an S1 nuclease-treated reaction and a matched buffer-only control. In the S1-treated reaction, 400 ng of gDNA was digested with 10 U of S1 nuclease (Thermo Fisher, EN0321) in the presence of 6 μ L of 5 $\times$ reaction buffer in a final volume of 30 μ L. Control reactions contained the same components but without S1 nuclease. Reactions were incubated at 37°C for 30 min and subsequently terminated by adding 2 μ L of 0.5 M EDTA followed by incubation at 70°C for 10 min. Following digestion, 1 μ L of each reaction was analyzed using an Agilent TapeStation with the Genomic DNA ScreenTape assay to quantify DNA fragmentation based on smear intensity. In parallel, 30 μ L of each reaction was separated on an agarose gel and visualized by SYBRTM Gold (Thermo Fisher, S11494) staining for 30 min. Smear intensity observed in the buffer-only control reflects pre-existing double-strand breaks, whereas the increase in smear intensity in S1-treated samples relative to controls indicates the presence of single-strand breaks, which are converted into double-strand breaks by S1 nuclease digestion.

Preprocessing of META-CS data

We preprocessed the META-CS data using a pipeline that expands the previously reported workflow⁴⁹ with additional features to accommodate both our modified experimental protocol and the somatic Indel calling method. First, paired-end reads were preprocessed by customized pre-meta which identifies Tn5 barcodes, merges overlapping read ends, and trims Illumina adapters. Then, BWA-MEM (v.0.7.17)⁸¹ was used to map reads to the human reference genome (GRCh37 with decoy) and generate a BAM file for each sample. Importantly, we then split the BAM file by unique pairs of barcodes which are used to distinguish original DNA fragments and help ensure that mutation calling was performed on a single-molecule level. Furthermore, given a relatively small pool of unique barcodes, barcode collision may occur where different DNA fragments are tagged by the same barcode pair around the same genomic region. This possibility also increases with our pooled-cell input with many more available DNA molecules at any locus. To address this challenge, we extracted Tn5 cut sites of each molecule (i.e. the start and end positions of each read pair) and used a combination of shared Tn5 barcode pair and Tn5 cut sites to identify original DNA fragments.

Somatic Indel calling from META-CS data

As the original META-CS method was developed for somatic SNVs, we established a somatic Indel calling pipeline (<https://github.com/gldong/duplex-indel>) that introduces modules to enhance the calling accuracy and remove false positives specific for Indels.⁸² First, we followed a workflow similar to SNV calling and piled up reads from the META-CS-barcode pair BAMs and a matching bulk WGS BAM to generate somatic Indel candidates by identifying variants that have at least 4 total non-reference (ALT) reads in META-CS with at least 2 ALT reads from each strand for duplex support, as well as no ALT read in bulk to remove potential germline variants.

When mate-pair reads from the same DNA molecule overlapped, the two mates were merged into a single read to avoid double counting. However, merging across repetitive DNA regions can introduce alignment artifacts and create spurious Indel evidence. To reduce these false positive calls, we defined a merging window as the overlapping region between mate-pair reads, and any Indel intersecting the merging window was discarded. When reads had different overlap boundaries, the largest merging window was used by taking the smallest start position and the largest end position.

In addition, we implemented a set of filters. A candidate site was filtered out if it met any of the following criteria, 1) overlapping with the low-quality regions as previously reported (um75- hs37d5.bed⁴⁹), 2) overlapping with gnomAD indels of $\geq 1\%$ population frequency, 3) located within 100 bp from another candidate site, 4) located within 10 bp of read ends, 5) having multiple Tn5 cut sites.

Single-strand Indel calling from META-CS data

Similar to double-strand Indel calling described above, single-strand Indel candidates were generated by piling up reads from the META-CS barcode-pair BAMs and a matching bulk WGS BAM, followed by identifying variants that have no ALT read from bulk and at least 8 total reads in META-CS with at least 4 reference allele (REF) reads from one strand (i.e. non-variant strand) and at least 4 ALT reads from the other strand (i.e. variant strand). In addition, candidates are required to have no ALT reads from the non-variant strand and no REF reads from the variant strand. Other filters are the same as double-strand calling above.

ALS and FTD snRNA-seq analysis

For snRNA-seq analysis on ALS and FTD, we used data generated for this study (i.e. in-house). We implemented the following pipeline to raw FASTQ files from both data sets. Cellranger (v.6.1.0)⁸³ with “--include-introns” was used to map reads to the human reference genome (GRCh37) and generate gene-by-cell count matrices. To remove potential ambient RNA contamination, we ran remove-background from CellBender (v.0.3.0)⁸⁴ on the raw count matrices with the default parameters to generate filtered count matrices for downstream analysis. For quality control, we filtered out cells with < 700 genes and mitochondrial gene percentage > 20%, which resulted in more than 65% cells removed in one sample (1700BA6-normal). So this sample was excluded

for further analysis. scDbfFinder⁸⁵ was used to remove doublets, and DropletQC⁸⁶ was used to calculate nuclear fractions where cells with a nuclear fraction < 0.25 were filtered out. Next, we used Seurat (v.5.1.0)⁸⁷ SCTransform v2 to perform normalization and variance stabilization followed by RunHarmony⁸⁸ to integrate and harmonize cell embeddings across all samples. The harmonized embeddings were used to perform dimension reduction (RunUMAP) and clustering using the Leiden algorithm (resolution = 0.8).⁸⁹ Cell types were manually annotated using previously reported markers⁹⁰ and canonical markers. Specifically, each marker gene's expression was visualized on the UMAP, and clusters with the highest and most specific expression were assigned to the corresponding cell type. A general cell type (excitatory neurons, inhibitory neurons, and non-neuronal cells) was annotated first, followed by subtyping of neurons. For example, clusters exhibiting high expression of SLC17A7 were annotated as excitatory neurons, and subclusters demonstrating concurrent high expression of CUX2 were further annotated as L2-3 IT.

Cell type fraction analysis in snRNA-seq

We used a Bayesian statistics framework^{91,92} to detect changes in cell type fractions in ALS and FTD compared to controls. This model implements Dirichlet multinomial modeling – Hamiltonian Monte Carlo (DMM-HMC) using R packages rstan (v.2.32.6) and bayestestR (v0.13.2) (<http://mc-stan.org/>). The input was a count matrix where each row is a sample and each column is a cell type. Counts in each row of the matrix follow a multinomial distribution where proportions of each cell type in the sample are modeled by Dirichlet. The prior probability of the proportions is another Dirichlet distribution with the fixed parameter $\alpha = 10^{-7}$ for equal expected fractions of cell types. We split samples into groups based on their disease status, brain region, and TDP-43 status, and calculated the posterior probability distribution (PPD) of each group. Then, we subtracted the PPD of control group from that of disease group to detect any significant changes in cell type fractions. Significance was determined if the log2 ratio of disease/control has 95% credible intervals above or below zero.

Differential gene expression analysis in snRNA-seq data

Differential gene expression analysis was performed using Nebula (v.1.5.3)⁹³ in R on in-house ALS and FTD data set and AD data set from SEA-AD consortium.⁹⁴ We downloaded the processed AD snRNA-seq data from SEA-AD consortium's web portal (sea-ad.org). For each data set, raw count matrices after preprocessing and cell type annotation were used as inputs and the design matrix was constructed with disease status as the predictor. Then, a negative binomial mixed model was built for each gene using individual as the random effect and total number of molecules in each cell as offsets. The output raw p-values were adjusted using the Benjamini-Hochberg method. Differentially expressed genes with FDR < 0.05 were considered significant.

qPCR analysis of RNASEH2 and RRM gene expression

Neuronal nuclei (2350 NeuN+ nuclei, ~5 μ L) from C9ORF72 FTD, AD and age-matched control brains were sorted into individual wells of 96-well plates with triplicates. Cell lysis, reverse transcription and qPCR were conducted using the Power SYBRTM Green Cells-to-CTTM Kit (Thermo Fisher 4402954) following the manufacturer's protocol. The following primers were used for qPCR analysis:

hRNASEH2A Forward - 5'- ACAGCCACTGGGCTTATACAG -3'
hRNASEH2A Reverse - 5'- TCCCTACGGTGTCCACGAATA -3'
hRNASEH2B Forward - 5'- TAACCCCTGTTCCAGGAGAAGG -3'
hRNASEH2B Reverse - 5'- ACACGTTATCCACCACAACCTTG -3'
hRNASEH2C Forward - 5'- AGGGACTCGAAGTGTCTGTTTC -3'
hRNASEH2C Reverse - 5'- TGTCACCATCACGTATCCAC -3'
hRRM1 Forward - 5'- GCCAGGATCGCTGTCTCTAAC -3'
hRRM1 Reverse - 5'- GAGAGTGTTTGCCATTATGTGGA -3'
hRRM2 Forward - 5'- ATTGGGCCTTGCGATGGATAG -3'
hRRM2 Reverse - 5'- GAGTCCTGGCATAAGACCTCT -3'
hRBFOX3 Forward - 5'- TCGTAGAGGGACGGAATTTGA -3'
hRBFOX3 Reverse - 5'- GCCGTTGGTGTAGGGGTTT -3'

TDP-43 knockdown in NGN2 neurons

CRISPR inhibition (CRISPRi) mediated knockdown of TDP-43 was performed in KOLF2.1J and FA11 iPSC lines harboring tetracycline-inducible Neurogenin2 (NGN2) transcription factor and dCAS9-ZIM3-KRAB-BFP. NGN2-mediated differentiation of iPSCs was achieved as previously described.⁹⁵ Briefly, undifferentiated iPSCs were maintained in StemFlex media (Gibco) with 1X StemFlex supplement (Gibco). On Day 0 of differentiation, cells were plated at a density of 50,000 cells/cm² in induction media (97% DMEM/F12 + 1% N2 supplement, 1% Non-Essential Amino Acids, 1% GlutaMAX, 2 μ g/mL doxycycline), and transduced with lentivirus expressing either non-targeting dual guide RNAs or TARDBP-targeting dual guide RNAs. Successfully transduced cells were selected with 5 μ g/mL puromycin for 48 hours. On Day 3 of differentiation, cells were re-plated on 0.1% poly-L-ornithine- and 5 μ g/mL laminin-coated plates in differentiation media (95% Neurobasal medium + 2% B27 supplement, 1% GlutaMAX, 50 mM sodium

chloride, 10 ng/mL NT3, 10 ng/mL BDNF, 10 ng/mL GDNF, 5 μ M aphidicolin, 2 μ g/mL doxycycline). Half-media changes were done every 3 days and cells were harvested on Day 10 of differentiation for either DNA extraction or immunofluorescence.

Immunofluorescence protocol

iPSC-derived neurons cultured for 10 days in 96-well plates were fixed in 4% paraformaldehyde for 20 minutes at room temperature. Neurons were permeabilized in 0.2% Triton X-100 diluted in PBS for 10 minutes followed by blocking in 5% bovine serum albumin diluted in PBS for 1 hour at room temperature. Neurons were incubated in primary antibodies diluted in 5% BSA for 2 hours at room temperature, followed by secondary antibodies diluted in 5% BSA for 1 hour at room temperature. Neurons were counterstained with DAPI and imaged at 40x magnification using a Nikon C1 confocal microscope. Primary antibodies: anti-TDP-43 (rabbit polyclonal, Proteintech 10782-2-AP, 1:500 concentration) and anti-MAP2 (chicken polyclonal, Thermo Fisher PA1-10005, 1:5000 concentration). Secondary antibodies: Donkey anti-rabbit Alexa Fluor 488 (Thermo Fisher A21206) and Donkey anti-chicken Alexa Fluor 647 (Thermo Fisher A78952).

QUANTIFICATION AND STATISTICAL ANALYSIS

All of the quantitative and statistical methods, strategies, and analyses are described in the relevant sections of the method details or in the table and figure legends.